The Clarity of Disclosure in Patents: An Economic Analysis Using Computational Linguistics

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Shupeng Sun

School of Economics and Finance

QUT Business School

Queensland University of Technology

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To my family

Abstract

Innovation is indisputably of great importance for economic growth. To achieve growth, investment in innovation must be incentivized, in addition, the knowledge created by innovation should be widely available for society. These two purposes can be achieved by a patent system. The motivation of the patent system is to ensure both aspects - the inventors receiving an exclusive right to recoup their investment on innovation and society receiving access to the new knowledge. However, extant research on patents has focused mostly on how patents can be used as a measure of innovation and how successful they have been used to incentivize innovation. There is very little empirical research studying the extent of disclosure that actually occurs in patent documents [1].

This thesis aims to explore and demonstrate the use of computational linguistic analysis to measure the readability of patent documents. Linguistic properties is an important dimension of disclosure, and it directly determine whether the readers of patents can reasonably extract the technological information contained in the patent documents. By using readability as a proxy for the extent of patent disclosure, I examine whether agents involved in innovation or technology commercialisation behave strategically in choosing different disclosure levels in patent applications or the choices of patent acquisitions, and whether the disclosure level may affect the probability of patents to be granted.

The first project studies whether applicants with different business models may strategically choose different disclosure levels when they draft patent applications. I choose

[1] There are many research focusing on the spillover effect of the patents by using patent citations[Almeida and Kogut, 1997, Bacchiocchi and Montobbio, 2009, Henderson et al., 1998, Jaffe et al., 1993, Sonn and Storper, 2008]. However, in this thesis, I aim to find out how readability can be used to measure the extent of disclosure of patent documents per se.

universities and practising entities as the starting point. Based on their business model, patents applied for by universities, which focus on generating income from licensing of innovations, should be more disclosing, that is more readable, than patents owned by private firms, which focus on in-house commercialisation. I find that universities, compared to practising firms, do choose higher disclosure levels for their patents.

After finding that applicants may tend to choose different disclosure levels at the stage of patent drafting, I further examine whether this strategic behaviour still exists in patent acquisitions. The second project choose the patent acquisition of non-practising entities as the research subject. Non-practising entities attempt to generate income from suing practising entities for patent infringement [Watkins, 2013]. Thus, one would expect non-practising entities to own, in particular, patents that are obscure and difficult to read, as this could cause confusion and lead to more unintended infringement [Bessen and Meurer, 2013]. To test this hypothesis I analyze the readability of patents which were acquired by non-practising entities, compared to those acquired by practising entities. However, I could not find that non-practising entities (patent trolls) choose to acquire less disclosing patents.

The third project studies whether disclosure may affect the patent examination. Recent empirical research has revealed that domestic patent applicants are significantly more successful than foreign applicants [Webster et al., 2014] in terms of passing patent examination. The reason(s) for this domestic inventor advantage is unknown and one possible explanation is that the language of international applications is not as clear as their domestic counterparts. I test this explanation and further study whether controlling for disclosure level can mitigate the domestic inventor advantage. The result shows that there is a positive relationship between disclosure level and granted rate, and the domestic inventor advantage which is found in Webster et al.'s study [2014] might be caused by language related factors.

Keywords

Patent disclosure; Readability; Computational linguistic analysis; Knowledge flows; Non-practicing entities; Patent acquisitions; National treatment; Domestic inventor advantage.

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Chapter 1

Introduction

Innovation benefits consumers with new and improved goods and services. Nearly all countries see innovation as essential to their economic development. However, technology is a "public good" [Posner and Landes, 2003], users cannot prevent simultaneous use by others (non-rivalrous), and it is very difficult for inventors to get paid from others' use (non-excludable). This creates a conflict with the need to encourage technology development. In order to solve this problem, the patent, a form of intellectual property, is used to turn the innovation output into the realm of property right [Barnes, 2011]. Patents have been considered as a tool to incentivize innovation for a long time [Jaffe and Lerner, 2003]. As far back as April 6, 1858, Abraham Lincoln, when he gave his first lecture on "Discoveries and Inventions" before the Young Men's Association of Bloomington, Illinois, said "The patent system. . .add the fuel of interest to the fire of genius, in the discovery and production of new and useful things." [1]

According to US patent law, the patentee can exclude others from making, using, offering for sale, selling or importing the invention throughout the United States, if the invention is a process, these rights extended to the products made by that process[2]. However, creating new knowledge is not sufficient for economic growth. To achieve economic growth, the new knowledge must be widely available for society. Giving inventors temporary monopoly rights can help innovators to recoup their investment and stimulate firms'

[1] See http://www.abrahamlincolnonline.org/lincoln/speeches/discoveries.htm
[2] See 35 U.S.C. §154: Article 154 of U.S. Patent Act

R&D activity. But it also prevents the diffusion of new knowledge. Therefore, apart from "incentivising innovation" (i.e., incentive function), the other fundamental purpose of patent system is "adding knowledge to the public domain" (i.e., disclosure function) [Roin, 2005]. To achieve this goal, the U.S. patent law requires that patent specifications shall be written in a full, clear, concise, and exact terms as to enable any person skilled in the art to carry out the invention[3].

The disclosure function is very important both theoretically and practically. In the economic model of the patent system, patent disclosure is assumed to play an important role in supporting following inventions [Denicolò and Franzoni, 2004, Scotchmer and Green, 1990]. The economic logic behind disclosure is very straightforward. A patent works as a contract between the inventor and the society in which the inventor is granted an exclusive right over the technology for 20 years[4] and the society benefits from disclosure of the technology so that other researchers can develop follow-up research based on it [Gallini, 2002, Scotchmer and Green, 1990]. Apart from offering access to new knowledge, disclosure function can also help to avoid duplicated research since researchers can learn what has already been done from patent documents and move to other areas which are less explored. Moreover, patent documents also signal technical information to potential investors which facilitates technology transfers [Fromer, 2008].

Disclosure function is very popular in U.S. courts, as the U.S. Supreme Court says that "the patent system represents a carefully crafted bargain that encourages both the creation and the public disclosure of new and useful advances in technology"[5]. In Bonito Boats, Inc. v. Thunder Craft Boats, Inc.[6], the Supreme Court even claimed that "the ultimate goal of the patent system is to bring new designs and technologies into the public domain through disclosure." The judicial doctrines highlight the practical importance of patent disclosure. For example, according to the written description requirements of Federal Circuit, patents can be held to be invalid if the patent documents do not sufficiently

[3]See 35 U.S.C. §112: Article 112 of U.S. Patent Act

[4]The patent term is different in different countries and for different patents. Take the US as example, for utility patents, the patent term is 20 years, but for some special patents, like pharmaceutical patents, the term may be longer than 20 years. See 35 U.S.C. §173

[5]See Pfaff v. Wells Elecs., Inc., 525 U.S. 55, 63 (1998)

[6]489 U.S. 141, 150-51 (1989)

disclose the technological information about the invention.

However, many scholars appear to be sceptical that disclosure has a strong effect in reality. For example, some scholars argue that the patent disclosures are often inadequate or unclear in reality (Devlin, 2010; Lemley, 2012; Roin, 2005). But most of these claims are based on anecdotal evidence[7]. Most of the research on patents focus on how patents can be used to measure innovation and how successful patents have been to incentivize innovation. The patent disclosure, however, has not been studied systematically, and there is very little empirical research studying the extent of disclosure that actually occurs in patent documents. Up to now, the most convincing evidence on disclosure function is from surveys. Inventors are asked questions about whether they prefer patents or other sources to get the technical information in the USA and Japan, and the results show that patents play a better role in knowledge diffusion in Japan [Cohen et al., 2002]. Walsh and Nagaoka [2009] use the same method and find that patent literature is more important for Japanese firms than American firms by surveying patentees in both of these two countries in 2007. Ouellette [2012] conducted a survey to study how researchers in the nanotechnology area use patents as a source of technical information and she finds that 64% of the 211 respondents have read patents for research purpose and 60% of those who read patents said they got useful information.

Given the importance of patent disclosure and the debate about it, together with the lack of empirical studies, it is worth studying how patent disclosure actually occurs in real world. The results from surveys can tell us how disclosure function works in reality, however, it cannot give us a general method which we can use to quantitatively measure disclosure in patent documents. In particular, survey results cannot tell the difference of disclosure level in individual patent document. Together with the other disadvantages of survey, such as lack of conscientious responses, and the difficulty to measure disclosure levels based on personal feeling, there is still a huge gap between the needs of quantitative analysis of patents disclosure and the extant research method.

[7]Devlin use several references to support his argument, but the references he uses are supported by other references. Both Lemley and Roin do not have empirical results to support their claims and Roin make his claim based on a speech of the CEO of a consulting firm specializing in intellectual property licensing.

In this thesis, I aim to explore and demonstrate the use of computational linguistic analysis to measure the readability of patent text and how readability can be used as a proxy of patent disclosure. As I said above, the linguistic property (readability) is one of the two most important dimensions of disclosure, and it directly determine whether the readers of patents can reasonably extract the technological information contained in the patent documents. Therefore, my assumption is that readability of patent documents is closely related to their disclosure levels, and I will test whether it is a good proxy in real data sets. By using readability as a proxy of disclosure level, we can get the disclosure level of an individual patent document, and with this we are able to answer questions, such as whether patent disclosure level is different between different types of applicants, or between different industries; what factors may affect the disclosure level of patent documents; whether the disclosure level may affect patent acquisition or patent examination. Besides, the use of readability measures in accounting and finance also give us a good (successful) precedent to introducing readability measures into patent documents. Readability determines whether readers can reasonably extract the information contained in documents [Loughran and McDonald, 2016]. Therefore, readability can be used as a rational proxy to study how the readability of financial reports (usually the annual reports: 10-K reports) may affect investors' behaviour, or be related to the firm performance [Lawrence, 2013, Li, 2008, Miller, 2010, You and Zhang, 2009].

In order to measure the readability of patent documents, a linguistic model from Vajjala and Meurers [2014c], which was shown to be one of the best non-commercial readability approach, is used in this thesis. The model contains 106 elaborately elected features and covers lexical richness, syntactic complexity, discourse and word characteristics of patent documents. With this method, I can have the value for all these 106 features for every patent document.

I begin with a study of whether applicants with different business models may strategically choose different disclosure levels when they draft their patent applications, implying that the different disclosing levels between patent documents are not random. I choose patents applied for by universities and practising firms as the starting point. Universities

and practising firms have different business models on their patenting (technology transfer vs in house commercialisation). Therefore, patents applied for by universities (with a focus on generating income from licensing of innovations) should be more disclosing, that is, referring to linguistic measures, more readable, than patents owned by practising firms (with a focus on in-house production plans), because a high disclosure level can make it easier to signal the technology contained in the patent and attract more potential investors. Together with the moral requirement of university research and poor performance of university technology transferring, it is reasonable to assume that universities may strategically choose a higher disclosing level for their patents when they draft them. I use all the patents applied for by universities and practising firms in the industry of nanotechnology (US Patent Classification No. 977), which were granted during 2009 to 2012, and apply the linguistic analysis model to the full text of these patents. The result shows that patents applied for by universities have higher disclosure level than those of practising firms.

After finding that applicants may strategically choose disclosure level for their patents, the natural question to ask is whether this difference matters. I then move to examine whether the disclosure level has also been considered in patent acquisitions. The second project chooses the patent acquisition of non-practising entities (hereafter shortened to NPE) as the research subject. NPEs acquire patents from other entities and attempt to generate income from suing or threatening to sue other practising entities (hereafter shortened to PE) for patent infringement [Cohen et al., 2014, Hagiu and Yoffie, 2013, Lemley and Shapiro, 2006, Reitzig et al., 2007, Watkins, 2013]. Thus, one would expect NPEs to own, in particular, patents that are obscure and difficult to read, as this could cause confusion and lead to more unintended infringement [Bessen and Meurer, 2013]. To test this hypothesis, I pick 100 patents acquired only by NPEs, and compare these to 100 patents acquired only by PEs. These two groups of patents are one-one matched basing on acquisition year, patent class, patent publication year, which mean the I match every patent that is acquired by NPEs with a patent from same/similar acquisition year, patent class, patent publication year that is acquired by PEs. However, I do not find disclosure level is significantly different between these two groups, which implies that

NPEs may not be able to discern the disclosure level of patents, and just buy every patent they can find on the market, or they acquire patents based on their technological value.

The third project studies whether readability affects the probability of patents to be granted. Recent empirical research has revealed that domestic patent applicants are significantly more successful than foreign applicants [Webster et al., 2014]. The reason(s) for this domestic inventor advantage is unknown, but one possible explanation is that the language of international applications is not as clear as their domestic counterparts. I choose 1000 patent applications which were published in January and February in 2010 from the pharmaceutical industry (US Patent Classification No.: 424). By using this dataset, I examine whether there is a significant difference between domestic and international applications in terms of readability and technological value, and whether better readability/technological value may lead to higher granted rate. The result shows that there is a positive relationship between readability and granted rate. And, the domestic inventor advantage which is found in Webster et al.'s study [2014] seems partially to be caused by language related factors.

To sum up, the main contribution of this thesis is that I firstly introduce the computational linguistic analysis to measure the readability of the patent text, and use readability as a proxy to study the extent of disclosure of patent documents, and its possible usage in research. Given the novelty of my methodology and research questions, this thesis develops a new perspective on the economic research of patents.

The thesis proceeds as follows. Chapter 2 gives the necessary legal and linguistic background information in this thesis, it includes a brief introduction of the patent system, the legal requirement of disclosure in the main patent authorities, economic and legal debates on disclosure function. In Chapter 3, I review the previous research on patents and patent statistics, as this will tell us what the fucus of research on economics of patents has been and what is the gap between the importance of patent disclosure and the extant research. I also review the the use of readability measures in accounting and finance, which supports the feasibility of introducing readability measures into the area of patents. This chapter tells us the novel methodology this thesis can bring to the current literature.

In Chapter 4, I give an introduction of the previous research on linguistic methods and the linguistic model I use in my thesis. Chapter 5 mainly focus on the first project, which is testing whether universities and practicing firms may strategically choose the disclosure level when drafting their patent applications. Chapter 6 studies the patent acquisitions of NPEs. I will test whether NPEs may intentionally choose patents which are unclear or less disclosing based on their notorious "nuisance value litigation" business model. In Chapter 7, I test whether domestic inventor advantage still exists after controlling the disclosure level of patent applications. Chapter 8 concludes and discusses the limitations of this thesis.

Chapter 2

Background

This chapter provides a basic introduction of the patent system, especially the prosecution of patent application and patent litigation, as this helps us to understand why patent disclosure is so important.

2.1 A Brief Introduction of Patents

Patents have a long history. The term "patent" is from the Latin "litterae patentes", which means an "open letter" used as proofs of rights and privileges conferred by medieval monarchs. The first industrial patent filling dates back to 1421, when Filippo Brunelleschi, a Florence architect "developed a crane system for shipping and transporting marble from the Carrara mountains". The modern patent law which aims to protect inventors from imitation rather than just a grant of a legal monopoly in a particular good, dates from the 18th century in Britain, the United States and France [Hall and Harhoff, 2012]. Right now, because of the increasing importance of innovation, around 190 countries all over the world have their own patent systems. The international conventions, including Paris Convention for the Protection of Industrial Property (Paris Convention), the Patent Cooperation Treaty (PCT) and TRIPS (The Agreement on Trade-Related Aspects of Intellectual Property Rights), together with other regional patent conventions (e.g. European Patent Convention) guarantee that patent law in different countries share some

basic features and the protection of foreign inventions.

Although patent systems are still different in different countries, the patent law harmonisation, especially the adoption of the TRIPS agreement in 1995, means the basic framework of patent laws from different countries is actually quite similar. In general, the patent system works as follows: To be granted a patent, an applicant needs to file an application which includes claims, specification and drawings (if necessary, it depends on the legal requirement in different countries). The government, normally the Patent (/Trademark) Office will examine the application and check whether the subject matter of an invention is patentable, and whether the invention is novel, non-obvious (or involving an inventive step) and susceptible of industrial application. If the application fulfills all these requirements, the Patent Office will grant patent right to the inventor. With a patent, the inventor can exclude others from using (including making, using, offering for sale, selling or importing) the invention, or license this technology to others. If the patent holder find that others have infringed his patent, he can sue them in court and ask for an injunction and damages. Usually he can file a lawsuit in a Federal District Court when a infringement happens [1]. A patent shall be presumed valid [2], and defendants can defend themselves by "unenforceability" and "invalidity"[3]. The normal remedies of patent infringement are "preliminary injunction" and damages. Apart from going to the final merit judgement, the defendant can choose to settle to get rid of a patent suit. In reality, most of the patent cases end by settlements, otherwise, the court will make a decision.

Since I use computational linguistic analysis to the full-text of patent documents, it is necessary to give a brief introduction of a patent document. A patent document usually consists of four parts: General information, Drawings, Specifications and Claims. General information provides the basic information about the patent, including patent serial number, title, the name of inventor or inventors together with the inventor's place of origin, patent number, patent classification, prior art, the examiner and patent attorney. The

[1] See 28 U.S.C. §1338(a)

[2] 35 U.S. Code §282

[3] Actually this "presumed to be valid" put NPEs into a advantageous position when they sue PEs on patent infringements, which I will discuss in my second project.

abstract of the patent is also included in this part. The second part is Drawings. Drawings are used to understand the technology contained in the patent, but they are not required for every patent. The third part is Specifications. Specifications are used to describe the claimed inventions in details, and it needs to fulfill the requirement of "enablement" and "best mode" which are discussed in the introduction above. The last part is Claims. Claims define the boundary of the patent right (the scope of legal protection). Claims are usually much shorter than specifications and the language is more formal and sometimes more obscure than everyday language [Durham, 2013].

2.2 Legal Requirements of Disclosure Function in Practice

The disclosure function is accepted by most of the patent systems as a main policy goal all over the world. USPTO (US Patent and Trademark Office) states that "patent is an intellectual property right granted by the USPTO . . . in exchange for public disclosure of the invention when the patent is granted". According to the 35 U.S.C. §112, the specification "shall contain a written description of the invention, and of the manner and process of making and using it, in such full, clear, concise, and exact terms as to enable any person skilled in the art to which it pertains, or with which it is most nearly connected, to make and use the same, and shall set forth the best mode contemplated by the inventor or joint inventors of carrying out the invention". These legal requirements are well known as (1) written description, (2) enablement, and (3) best mode.

Many other countries or authorities have similar legal requirements, such as:

TRIPS: Article 29 requires that members shall require that an applicant shall disclose the invention in a manner sufficiently **clear** and **complete** for the complete for the invention to be carried out by a person skilled in the art and may require the applicant to indicate the **best mode** for carrying out the invention known to the inventor at the filing date or, where priority is claimed, at the priority date of the application.

EPO (European Patent Office): According to Article 83 of European Patent Convention, the European patent application shall disclose the invention in a manner sufficiently

clear and **complete** for it to be carried out by a person skilled in the art.

Australia: Section 40 of Australia Patent Act (1990) requires that a **complete** specification must disclose the invention in a manner which is **clear** enough and **complete** enough for the invention to be performed by a person skilled in the relevant art, and the patent claims need to be **clear** and **succinct** and fairly based on the matter described in the specification. These are known as the "sufficiency requirement" and "fair basis" requirement.

These rules all require that inventors should disclose patents in a "clear", "complete" and "full" manner.

The disclosure requirement in U.S. patent law has three separate components: enablement, written description, best mode[4]. Enablement requirement means that the patent must describe the invention in such clear and exact terms that persons skilled in the art can understand how to make and use the invention without "undue experimentation". Written description requirement means that the specification of patent must use full, clear, concise, and exact terms to explain the manner of making and using the invention. The best mode requirement means the specification must set forth the best mode contemplated by the inventor for carrying out his invention. Failure to fulfill the enablement and written description requirement would lead to the patent claim being canceled or held invalid[5].

2.3 Economic and Legal Debates on Disclosure Function

Given that many scholars argue that the society actually receives little benefit from the disclosure function, here I examine their arguments in turn and prove that disclosure is still a compelling justification for the patent system.

The first critique is that many inventors file for patents only if their inventions are "self-disclosure" (the most popular example of "self-disclosure" patent is the paper clip patent) or easy to reserve-engineer, otherwise, they will never ask for patent protection

[4] 35 U.S.C. §112

[5] Failure to disclose best mode is no longer a basis to cancel or invalid a patent claim, according to America Invents Act (2011)

[Devlin, 2010, Roin, 2005]. It is true that inventors may decide to keep the inventions secret if their inventions are not transparent to the world [Lemley, 2004]. In practice, patents are indeed not the only method which innovators could use to protect their inventions. For example, trade secret is also widely used by many firms, and one of the most famous examples is the secret Coca-Cola formula. However, the patent system now still plays a key role in the process of innovation [Jaffe and Lerner, 2011]. For firms, patents and patent strategies are still a very important part of firms' competitive strategies [Paik and Zhu, 2013]. In 2013, patent applications filed worldwide amounted to around 2.57 million, up 9% from 2012 (WIPO). For all these 2.57 million patent applications, it is impossible to check whether they are all easy to reserve-engineer or not. In extreme cases, even if we assume that all these patents are self-disclosed or easy to be reverse-engineered, patent disclosure is still necessary for disseminating technical information because reverse engineering all these patented technology is costly, not to mention that patent disclosure can help reduce wasteful duplicate research and facilitate following invention.

Second, based on the "wilful infringement rule" (used in the U.S. Federal Circuit), some scholars believe that inventors often do not read patent documents, because intentional infringement would lead to higher damage or punishment compared to the tort with fault [Lemley, 2012, Lemley and Tangri, 2003, Lichtman, 2005, Roin, 2005][6]. For example, Lemley [2012] claims that lawyers often advise researchers not read competitors' patents in order to avoid wilful infringement. Roin [2005] also points out that many researchers do not treat patents as a source of technical information, they only use patents when searching for potential infringement problems or are simply not reading patents. However, there is no empirical evidence so far that supports for any of the claims above. On the other hand, based on the survey reported by Ouellette [2012] on nanotechnology industry, researchers actually do not avoid reading patents because of legal concerns like wilful infringement. When asked whether they "worry that reading patents could have negative legal effects", only 6 out of 134 participants who read patents (4%) said yes, and only 3 out of these 6 were concerned about liability for infringement. Therefore, it seems

[6]If willful infringement is confirmed by the jury, the court can increase the damages awards by up to three times. 35 U.S.C. §284.

that the effect of the "wilful infringement rule" is a bit overstated, or at least based on no empirical evidence.

Third, researchers also claim that the language of patent documents is too difficult or unclear which means the payoff of reading patent documents is often dubious [Devlin, 2010, Lemley, 2012, Roin, 2005]. It can be a big problem if it is true. Intellectual property was created to protect inventors' investments. As a form of property, patents must be able to serve the "notice function" which is considered as a basic function of a property. It means that patents must be able to clearly tell non-owners the boundaries of patents. Otherwise patents cannot be enforced efficiently or the cost of enforcement will be very high. Of course it is true that it is the claims, not specification, which decide the scope of protection, but claims should be based on the specifications. If the specifications are unclear or difficult to understand, it will be difficult to clearly understand patent claims and cause a lot of wasteful repeated research and unwilful infringement. However, as we discussed in the introduction, most of these claims are based on anecdotal evidence, and there is no empirical research which can clearly tell the disclosure level of patent documents that actually occur in reality.

Based on the previous analysis, we may find that these critiques on disclosure function are not that convincing or at least lack empirical support. To study whether patent disclosure function is one compelling justification of the patent system, we need first measure the patent disclosure that happened in the real world, and how patent disclosure can affect or be affected. This is the focus of this thesis.

Chapter 3

Literature Review

The literature for this thesis is from two areas: the research on patents and patent statistics and the research on linguistic methods.

3.1 Previous Research on Patents

With the development of knowledge economy, the large increase in research on patents give us voluminous literature resources. However, most of the research focus on how patents can be used to measure innovation and how successful patents have been to incentivize innovation.

Innovation is extremely important for economic growth and, as early as the 1950s, Solow [1957] had already proved the essential role of technical change in economic development. The gross output per man doubled during 1909 - 1949 in the US, and 87.5% of the increase is associated with technical change, much more than the increase bought by increased use of capital (12.5%). Because of its importance, various attempts have been made to measure innovation. Initially, the analysis has focused on Research and Development (R&D) expenditures. But these measures have already been proven to be highly imperfect[1], and patent data seems a better choice [Lerner and Seru, 2015], since

[1]First, R&D can only measure the material input; second, RD expenditures are input of innovation, and innovative output may vary tremendously because of the difference of effectiveness.

patents are the only indication of innovation which covers almost every industry in most of countries and over long periods of time [Trajtenberg, 1990].

Initially, patent counts, that is the number of patents issued in different countries, to different types of applicants and at different times, used to be the major information economists focus on to measure the innovation [Griliches, 1990]. However, patent counts provide only a very noisy measure of innovation, because patents vary enormously in their value or importance [Trajtenberg, 1990]. In the literature, this question as well as the question of whether patents differ in the type of innovation (important vs trivial or fundamental vs applicative), has been addressed using patent citation on its own or as a weight of patent counts to measure innovation [Carpenter et al., 1981, Hagedoorn and Cloodt, 2003, Trajtenberg, 1990]. Carpenter et al. [1981] find that the average number of citations received by patents which are associated with important inventions (the products received IR 100 award established by the journal Industrial Research and Development) are significantly higher than randomly selected patents. This result proves the positive relationship between the number of citations and the value of innovation.

After that, Trajtenberg [1990] proves patent counts weighted by citations can overcome the imperfection of simple patent counts and be used as an effective indicator of the value of innovations. By using a simple weighting scheme, Trajtenberg weighs each patent in Computed Tomography by the actual number of forward citations, and finds that the new index can be used as indicative of the value of innovation[2]. In this thesis, patent citation, together with other indices of patent value, is used to control for the innovativeness of patents. In this thesis, I also rely on patent citation data, together with other indices of patent value, to control for the innovativeness of innovations.

For how successful patents can incentivize innovation, Nordhaus [1969] finds that stronger patents will obviously induce more investment in R&D, when referring to a single and isolated invention[3]. However, the new development in innovation process,

[2]The new index for a single patent equals the sum of its forward citation in year t plus 1. Then the yearly index in certain industry or area naturally become the sum of this new index. This combination of simple patent counts and patent forward citations is better to display the yearly innovation output than relying on simple patent counts alone.

[3]One product only covered by a single patent, and this patent is not built on precedent patent(s).

such as fragmented invention and cumulative invention[4], makes the incentive function which works quite well in single-inventor model questionable.

In the case of cumulative invention, it is extremely difficult to balance the incentive between the first inventor and the following inventors. A broad protection for the first inventor would reduce the incentive of following inventors to develop the second-generation products, since they cannot market their products without licensing from the first inventor and this puts them in a very weak bargaining position. On the other hand, a narrow protection for the first inventor cannot sufficiently reward the first inventor, and their incentive to innovation becomes weak. Scotchmer [1991] points out the prior agreements or collusive licensing may help to balance this dilemma. However, these solutions may not be allowed by patent authorities. In addition, they seem impossible to implement, since it is extremely hard for the first inventor to identify the second inventor. Gallini and Scotchmer [2002] also point out that contracting can help to mitigate this imbalance, but contracting may lead to antitrust concerns.

For fragmented inventions, Hunt [2006] finds that cumulative innovation is more likely to occur in high-tech industries, including electronics, computers, and semiconductors, and it further finds that if patent overlap is strong enough, the patent will reduce rather than increase the R&D investment. On the other hand, Galasso and Schankerman [2010] find that the establishment of a Court of Appeals for the Federal Circuit is correlated with the reduction of patent disputes duration. This is because the outcomes of disputes will be more certain when going to a centralized court.

The other prominent use of patent citations is tracing flows of knowledge. In their seminal paper, Jaffe et al. [1993] use patent citations as the "paper trail" to measure the knowledge flow. They use patent applications from universities, large firms and other firms[5] in 1975 and 1980 as the originating patents, and identify all their citations[6]. They

[4]Fragmented invention means a single product covered by more than one patents which are owned by different holders. Cumulative invention means one patent is built on anther patent.

[5]The large firms are the 200 top-R&D-performing firms in the United States, and other firms are firms that are not in the top 200 list.

[6]The 1975 originating patents have 950 patents and had received 4750 citations up to the end of 1989. The 1980 originating patents have 1450 patents and had received 5200 citations by the same time

then match every citing patent (the patent citation) with a control patent which has the same application year and is in the same patent class to control the distribution of time and technological areas. They then compare the probability that citing patents are geographically proximate to the cited patents, with the probability that control patents to the cited patents. They find that citing patents are more likely to be geographically approximate with originating patents. The follow-up research addresses knowledge flows based on their methodology. Almeida and Kogut [1997] find that small firms prefer to avoid doing innovation in industries that are well-established by large firms, and are integrated into the local knowledge network with a higher level than large firms. Many may argue that this localization may fade and disappear eventually with the rapid development of telecommunication. However, Sonn and Storper [2008] study 2082226 patents which were applied for by U.S. inventors from 1975 to 1997, and find that the percentage of local citations has increased over time, even after controlling for the current distribution of knowledge-production activities. Scholars also estimate whether there is any difference of knowledge diffusion from patents which were applied for by different types of applicants (university, public laboratories and corporate patents). By using data from EPO which covers different countries and different industries[7], Bacchiocchi and Montobbio [2009] study the different characteristics of knowledge flows in different countries and industries, and they find university patents in the U.S. and in the Chemical, Drugs and Medical industries are more important than corporation patents, and patents which were applied for by universities and public research organizations diffuse more rapidly. Someone may argue that the better performance on patents of US universities is due to the introduction of Bayh-Dole Act which tries to stimulate university patenting and technology transferring. However, this better performance existed even before the Bayh-Dole Act, not to mention that the effect of the Bayh-Dole Act is still controversial Henderson et al. [1998].

Comparing to the research on the above area, although the effect that patent documents can facilitate following on inventions has been widely accepted, disclosure function has not been systematically studied. As early as 1950, Machlup and Penrose [1950] state

[7]Their dataset contains 793040 patents from six countries (US., UK, France, Germany, Italy and Japan) during 1978 to 1997, these patents are all from six industries: chemicals, computer&communications, drugs&medical, electricalelectronics, mechanical and others.

that the patent is a bargain between society and the inventors, which can incentivize the inventors to disclose their "secret" to exchange for the protection of their "secret" from the state. This argument is repeated by other scholars [Kitch, 1977, Posner and Landes, 2003], and patent disclosure is assumed to have a positive effect on following inventions [Denicolò and Franzoni, 2004, Scotchmer and Green, 1990]. On the contrary, as I discussed in the section Background, some scholars suspect the effect of disclosure. However, since there is very little empirical research studying the extent of disclosure that actually occurs in patent documents, neither of these two arguments can be confirmed.

So far, most of the empirical evidence about patent disclosure is from surveys. The earliest dates back to 1980. To review the patent system in Australia, large numbers of patent attorney clients, practicing firms, professional engineers, individual inventors and government departments were asked their opinion about the use of patent information in Australia by surveys [Mandeville, 1983]. This survey shows that "Check potential patent infringement" is the most important reason for use of patent information (57% for small firms, 51% for medium firms and 44% for large firms) for firms. And access state of art only accounts from 7% to 18%. And there are only 4% of firms use published patent specifications as the main sources of technological information, compare to 21% for technical and trade journal and around 24% for overseas visits.

After that, a survey was carried out on UK-based small and medium-sized firms (SMEs) to collect information about how they use patents and patent information. This showed that 55.4% of them search patents for commercial reasons, and 40.7% do so for both technical and commercial reasons, while only 3.9% search patents only for technical reasons. This indicates that SMEs are very unlikely to search patents for technical reasons unless they want to do so for commercial ' as well.

In 1998, Jaffe et al. [2000] conducted a creative survey to study the communication between patents and their patent citations. They randomly choose about 600 citing inventors from patents which were granted in 1993, and these inventors were asked about three patents (two are the patents they cited in their patents, and the other one is a "placebo" which is similar but actually they did not cite). At the same time, one of the two cited

inventors was also surveyed. Based on the survey data, they found the existence of significant communication between inventors, which indicates the knowledge flows from cited inventors to citing inventors.

Based on a survey of managers of R&D departments of practicing firms in the US and Japan, referring to patents as a channel of information flow, Cohen et al. [2002] study the different R&D knowledge flows and spillovers in these two countries. They find that patents as a source of information are more important in Japan (61.8%) than in the US (49.1%). On the other hand, more Japanese firms choose "information disclosure" as the reason for not applying for a patent than their American counterparts (46% vs 26%). The findings above might be able to indicate that patents are a more important source of knowledge in Japan than in the US. They also mentioned that this difference might be due to the patent system, including rules of patent priority, disclosure and patent opposition procedure.

The most recent impressive survey on disclosure function of patents is conducted by Ouellette [2012] in 2010. To study how patents can be used as a source of technical information in the nanotechnology industry, 1078 researchers were emailed the survey and finally 211 responses were received. In all,, 64% of respondents have read patents before, and 62% of those who have read patent(s) aim to check their research's patentability, 70% (44% of the total respondents) were looking for useful technical information, and 40% for getting an idea about how other researchers solve a certain problem. Based on her study, reading patents is still a popular way to obtain information among researchers in the nanotechnology industry. Another interesting finding is reading patents is more common among industry researchers and experimentalists than basic researchers.

From all the excellent research above, we can see how patents have been used, especially on disclosure function. We find that almost all of the empirical evidence about patent disclosure is from surveys. These surveys tell us how patents are used as a source of technical information in the real world. However, as discussed above, the initial generation of knowledge is taken as given and not analyzed, and the extent of disclosure on individual levels is not studied as well. For the questions such as whether patent

disclosure is different from between different types of applicants, in different industries, etc, the role of disclosure level in patent acquisitions and patent examination, have not yet been studied. Given the bitter debate on patent disclosure function, my research can address this crucial gap by developing a new methodology for measuring the extent of disclosure in the patents as well as shedding light on further research or policy adjustment.

3.2 The Use of Linguistic Measures in Finance and Accounting

Computational linguistics has not been widely used in research in patents, the only example is [Packalen and Bhattacharya, 2015]. To study which type of work (new idea based or old idea based) is more likely to stimulate follow-up innovation, they use words and word sequences which firstly describe a certain technical term as the concept, and count the number of patents that use these concepts. They find inventions based on new ideas are more likely to stimulate follow-up inventions than inventions based on old ideas. However, Packalen and Bhattacharya still focus on the technologies that patents contain, and linguistic methods are used to pick up the technical terms rather than measure the disclosure level of patent documents. Another interesting study is from Younge and Kuhn [2016]. They use a machine-automated method, a more advanced technique to measure the patent-to-patent similarity to improve the existing patent classification systems. However, their study still focus on the measure of technological similarity.

Unlike in economics, linguistic measures have been widely used in accounting and finance. This may partially attribute to SEC'S plain English initiative of 1998. Improved computer technology which makes massive text analysis available also helps the increase in using of textual analysis in accounting and finance [Loughran and McDonald, 2014].

Antweiler and Frank [2004] firstly study whether internet stock messages can affect market volatility. They use computational linguistics methods to measure the bullishness which was described in more than 1.5 million messages posted on Yahoo! Finance and Raging Bull about 45 firms in the Dow Jones Industrial Average and the Dow Jones Internet Index. They use Naive Bayes algorithm and Support Vector Machine (another

algorithm) to classify documents into three groups: buy, hold, and sell, and how this information can be used to predict market volatility. They find that a positive shock predicts negative returns on the next day and greater disagreement between message boards actually predicts fewer rather than more trades on the next day.

Following Antweiler and Frank [2004], other researchers have used this method on other forms of financial documents. Tetlock [2007] tries to measure the relationship between the stock market performance and media content (daily content from "Abreast of the Market" column of Wall Street Journal). By using General Inquirer (GI) to convert the column into numeric values, he finds the "tone" of content can predict the stock price, that is a high media pessimism can predict downward pressure on market prices, and unusually high or low pessimism can predict high market trade volume.

After that, Tetlock et al. extend the linguistic analysis to all Wall Street Journal (WSJ) and Dow Jones News Service (DJNS) stories about individual firms (S&P 500 firms from 1980 to 2004), and study how the negative words used in these stories can predict firm's earnings and returns [Tetlock et al., 2008]. By using negative words fraction in these stories, they find that negative media coverage, especially those about fundamentals, can predict low firm earnings. Similar research on the effect of media coverage can also be found in Dougal et al. [2012].

The linguistic analysis used in the research above still focus on the content of financial documents, the bullishness level of message, and the negative words in the column and news story are related to the content of documents. In 2008, Li firstly introduce readability to assess financial documents. Based on "management obfuscation hypothesis"[8] [Bloomfield, 2002], he tests the relationship between the readability of annual reports and firm performance. Li uses the Fog Index and the simple word count as the indices of readability, and he finds that firms with lower earnings tend to have harder reading annual reports (higher Fog index value and longer report), which supports "management obfuscation hypothesis" based on empirical evidence.

[8]Managers may try to obfuscate their financial documents when firms have a bad performance, if this can reduce the market reaction because of the hardness of extracting information from less readable financial documents.

After that, Miller [2010] examines about 13,000 10-K reports (annual report) between 1995 and 2006 to study the relationship between readability and trading volume. By using Fog Index and a unique index called READ_PE which is designed by the author based on the specific characteristics of financial documents, he finds that small investors tend to reduce their trading volume with the complexity of 10-K reports, and this effect is not significant for large investors. The possible reason for this effect is the assumption that within the relatively short window surrounding the filing date, it is more difficult for small investors to process the difficult reports than large investors.

Lawrence [2013] studies the relationship between individual investment and financial disclosures. As with Li, Lawrence uses Fog index and length of annual reports as the indices of readability. Based on the analysis on trades and portfolio of individual investor between 1994 and 1996, he finds that individual investors tend to invest more on the firms with "clear" and "concise" 10-K reports, and the investment return increases with higher readability level.

The readability level can also affect the trading volume. Franco et al. [2015] analyze the readability level of equity analysts' reports from Investext during 2002 to 2009 (using a combination of three measures: the Fog index, the Flesch-Kincaid index and the Flesch Reading Ease Index), and they find a positive relationship between readability level of analyst reports and trading volume.

Given the limited scope of this thesis, I cannot discuss all the text-based research in accounting and finance. However, all the research above, especially those on readability of financial reports and how this may affect investors' trading behavior, gives a good precedent of how computational linguistic method can be used in different area and how readability of documents (not just the content itself) may affect people's behavior.

Chapter 4

Linguistic Method

This chapter provides a basic introduction of linguistic method, and the linguistic model used in this thesis.

4.1 The Previous Research on Linguistic Method

The study of linguistics has a long history of research on readability [DuBay, 2007]. Possible applications of this readability assessment include evaluating the textbooks of language learners, providing reading materials suitable for students to read and so on. The traditional readability formulas, such as the Flesch-Kincaid formula [Kincaid et al., 1975], were based on shallow, easy to count features such as the average sentence length and average word length. Some formulas also integrate counts of words on specific word lists [Dale and Chall, 1948]. The traditional readability measures are essentially based on rough generalizations (long words are rare, long sentences are difficult) and provide only a quantitative measure, not an actual characterization of the different aspects of language determining readability. Recently, computational linguistic approaches have started exploring alternatives. Current approaches, inspired by research in psychology, emphasize the role of coherence/cohesion for comprehension (e.g.,Graesser et al. [2014]), and accordingly look at lexical and grammatical features [Callan and Eskenazi, 2007, Lu, 2010, Schwarm and Ostendorf, 2005]. The ability of current supervised machine learning

algorithms capable of handling large feature spaces and feature sets not satisfying strict independence assumptions makes it possible to explore the impact of a wider range of linguistically and cognitively grounded features with potential impact on readability.

For the present research, the most important features of the approach are measures of linguistic complexity developed in Second Language Acquisition (SLA) research. The notion of Complexity in SLA, since Skehan [1991], is considered as one of the three principal dimensions of proficiency in the triad Complexity, Accuracy, and Fluency [Housen and Kuiken, 2009]. The research since 1990s on complexity in SLA has already established a rich inventory of empirically observable language properties that can be connected to the development of language proficiency and interpreted in terms of complexity [Lu, 2010, 2011, 2012].

Apart from complexity features, Propositional Idea Density (PID) [Brown et al., 2008, Kintsch and Keenan, 1973, Turner and Greene, 1977], as a meaning-based notion can provide a good complement focusing more on characteristics of language form at the sentence level and above. The human sentence processing literature can also offer two further types of potentially valuable characterizations of complexity. On the one hand, psycholinguistic proposals such as the Dependency Locality Theory [Gibson, 2000] provide a cognitive motivation for complexity features encoding dependency lengths in a sentence. On the other hand, one can compute more process-oriented measures such as *surprisal* to reflecting human sentence parsing cost Boston et al. [2008], and how it can predict reading difficulty. Finally, complexity measures encoding language use, as, for example, recorded in representative corpora such as SUBTLEX Van Heuven et al. [2014], provide us with the opportunity to operationalize the elaborateness in the lexical domain in terms of the frequency of occurrence. Recently created, wide-coverage age-of-acquisition norms Kuperman et al. [2012] also provide information on potentially relevant characteristics related to lexical experience.

Based on these useful strands of research, the readability assessment method used in this thesis, which is from Vajjala and Meurers [2014c], derives a large set of features from these research, and is one of the best non-commercial readability algorithm for English

in different contexts [Vajjala and Meurers, 2012, 2013, 2014a,b]. It has shown that this rich complexity feature set supports a reliable readability assessment at the sentence level [Vajjala and Meurers, 2014c].

4.2 Our Linguistic Model

The linguistic model used in this thesis integrates a wide range of linguistically and cognitively grounded features on readability. In addition, Vajjala and Meurers have already proven that this model was equally successful in different genres and domains. Therefore, in sum, the model provides a fine-grained set of observations characterizing readability, from aspects of form (how complicated is something expressed in grammar and lexis) via aspects of meaning (how much content is being conveyed) to aspects of cognitive processing (how ambiguous is something at a given point in the sentence) and language use (how common are the words and expressions used).

In detail, there are 106 features in the linguistic model and all of these 106 features can be broadly classified into four categories: lexical richness and POS (parts of speech) features, syntactic complexity features, discourse features and word characteristics features.

The features of lexical richness are from Second Language Acquisition (SLA) research, including the type-token ratio, corrected type-token ratio and lexical variation (noun, verb, adjective, adverb and modifier variation). These features also include the density of different POS in the texts and study the relation of POS density with the overall score.

The features of syntactic complexity focus on the structure of the sentences. Features in this category are from SLA research and parse-tree based features. Berkeley Parser [Petrov and Klein, 2007] is used to generate parse trees and Tregex pattern matcher [Levy and Andrew, 2006] is used to count the occurrence of various syntactic patterns. Features like average lengths of various production units, measures of coordination and sub-ordination, number of phrases of various categories, average lengths of phrases, parse

tree[1] height are used in this group.

Word characteristics focus on the readability of words. Morpho-syntactic, psycholinguistic, semantic characteristics and information on their age-of-acquisition are used for readability assessment. These features, including word familiarity, age of acquisition, proportion of countable nouns and word concreteness and so on, are constructed based on a new database with age-of-acquisition norms for English words and using Wordnet10, a lexical database for English.

Discourse features look at relationships between sentences and model discourse cohesion at a very shallow level. Cohesion is a characteristic of the text. Cohesion is an objective property of the explicit language and text. There are explicit features, words, phrases, or sentences that guide the readers in interpreting the substantive ideas in the text, in connecting ideas with other ideas, and in connecting ideas to higher level global units (e.g., topics and themes).

Analyzing patent texts using the computational linguistic analysis model capturing language characteristics related to linguistic complexity and readability makes it possible to objectively identify and compare patent texts and correlate the fine grained readability fingerprints with language-external labels, such as whether patents applied for by universities are more readable than those filed by practicing firms, and if that is the case, in which respects (study 1), whether the linguistic characteristics can identify non-practicing entities' patent acquisitions (study 2), and whether domestic and international applications are equally successful when using comparable language (study 3). The question of which complexity features are informative with respect to these distinctions from the patent domain, and how they are best aggregated and weighted, provide clear opportunity for fruitful interdisciplinary collaboration and further development of the computational linguistic techniques.

[1]In linguistics and natural language processing, parse tree is the labeled tree structures to phrases and sentences of human languages.

Chapter 5

Are University Patents more disclosing than Those of Corporation? First Results

5.1 Introduction

In this chapter, I aim to explore whether different subjects (with different incentives) may strategically choose different levels of disclosure. I choose patents that have been applied for by universities and practicing firms as the starting point. My assumption is that universities and practicing firms serve different business models on patenting, and this difference would affect the disclosure level of their patent applications.

The main purposes of universities are teaching and research, and the dominating business model for university technology transfer is licensing patents to the highest bidder [Valdivia, 2013]. On the contrary, practising firms normally seek to commercialise their R&D results by themselves and try to maximise their profits. In order to attract potential investors, universities would describe their inventions relatively more clearly in patent specifications[1]. Compared to universities, practising firms are more likely to be reluctant to clearly disclose their technical information. This is because patent disclosure can facilitate "invent around", and this is considered as "a limitation on the monopoly power" of the patent right [Posner and Landes, 2003]. In reality, firms may only disclose their

[1]As we discussed in the introduction, one of the effects of disclosure function is signalling technical information contained in patents and facilitating technology transfers

technical information for a defensive purpose, for example, disclose some key information to the public (enlarge the prior art) to make it more difficult for their competitors to apply for patents in related area [Baker and Mezzetti, 2005].

The moral requirement of university research also makes universities less likely to block and prevent knowledge flows. In 2007, several universities, including Caltech, Stanford, MIT, Harvard, signed a statement in which they all promised to be mindful of public interests and declared that "exclusive licenses should be structured in a manner that encourages technology development and use"[2].

Besides, the Bayh-Dole act, which created the legal framework in the U.S. for universities to own patents on publicly funded research (which is almost all of their patents) explicitly makes realization of the economic and social benefits of the invention a goal of the law. Hence universities are legally mandated to foster the diffusion of their patents[Henderson et al., 1998].

Furthermore, the situation of university research would push or force universities to pay more attention on technology transfer. In 2012, 130 of 155 American universities did not generate enough licensing income to cover the wages of their technology transfer office (TTO) staffs and the legal costs for the patents they filed. Around 84% of universities operating technology transfer were in the red, and over the last 20 years, on average, 87% were in the red [Valdivia, 2013]. Besides, only about 5% of patents are licensed at most of the universities in America, the rest just become the burden of TTO due to required maintenance and legal fees [Ledford, 2013]. In this case, in order to attract potential investors, it is safe to assume that universities would like to signal more technical information in their patent applications.

An extant empirical research also supports my assumption about universities' patenting motivation. In their empirical study about patent filing motives of individual inventors, universities, small companies and large firms, Veer and Jell [2012] use data from a survey among applicants at the European Patent Office. Six hundred and forty-one randomly chosen applicants were asked to rate their incentives on preventing imitation, signaling

[2]See https://otl.stanford.edu/documents/whitepaper-10.pdf

technical information, licensing and blocking and so on. In this survey, the most striking result is universities' motive of licensing, with 33% of universities rating licensing as the sole important patenting motive while only 4% of other applicant types have the same choice. For the prevention of imitation, most applicant types rate this as the most important motive of patenting, but only universities rate this as less important (only 20% of universities rate this as important or very important).

Taking these factors into consideration, it is reasonable to conjecture that universities have more incentive to write clear and comprehensible patent specifications than practicing firm applicants. I develop a simple theoretical model of the patenting behavior of universities and practicing firms and I also test how the different incentives of universities and practicing firms would affect the disclosure level of their patent documents based on the linguistic analysis results on patent samples of universities and practicing firms.

5.2 The Model

The standard way to analyze incentives for innovation in the patent system is to consider investments and returns of research and development investments. To provide a framework for the following discussion, I introduce a simple theoretical model by using the disclosure as a variable. I abstract from the process of innovation and focus on patent application and use of the patents. Thus I consider two stages. First, the application process: a game between the inventor or applicant who may act on behalf a firm, or a university, and the patent office or examiner. Second, the market game of the inventor or their organization. With the latter I allow for organizations that can directly use the technology to supply the market. I refer to these organizations as organizations with the capacity to use the invented technology in-house (practicing firms); as well as for organizations (universities) that need to license the technology to firms using the technology to provide services to the market.

Consider the first stage, the economy I consider in my model has a large number of agents-different organizations like firms or research institutions, as well as individual

inventors-generating new technologies and a large number of end-users-firms, that may or may not be active in research and development-of these inventions.

With respect to their innovation technology, the inventing agents serving a specific market are identical. But agents may differ in their capacity or intention to commercialize a product themselves − to allow for example for universities of other non-practicing entities. In this case each innovation can be treated as an individual incidence, and given the large number of firms, there is no reason to delay an application for strategic reasons. I assume that innovations are original and are characterized by the size Δ of the inventive step which is randomly drawn from a distribution F(Δ). The firm has to decide whether to patent this innovation, by investing a fixed amount and choosing a level of disclosure *r* (for level of revelation). The patent office grants a patent if $\Delta>0$[3]and the level of disclosure is larger than a critical level *r**. This patent examination process may be probabilistic, but for now let us assume it is deterministic and free of error.

Once this decision is made, we enter stage 2 of our model. If the firm chose not to apply for the patent or the patent was not granted, without loss of generality we can normalize the additional value for the firm to 0, that is, I assume that the technology is quickly dissipated in the market and a firm cannot claim any additional value from that innovation. If a patent is granted and is put to use the technology generates an additional value to the company of V(Δ, r).

I assume $\partial V/\partial\Delta >0$ and $\partial V/\partial r <0$, that is, the value increases in the size of inventive step but decreases in the level of disclosure chosen. The latter assumption captures the idea that higher degrees of disclosure may lead to higher spillovers or quicker catch-up of competitors circumventing the patented technology. For a firm with in-house capacity to use the patent, this value is straightforward to capture. For a non-practicing entity, like a university, the technology needs to be licensed to a producer. For simplicity of this simple model/framework I assume that markets are competitive, in the sense that if an inventor finds producers who are able to understand the technology to its full extent, he will always find at least two so that the inventor can achieve V(Δ, r). Whether such

[3]Here I just simplify the real examination on patent novelty and non-obviousness: as long as the size of novelty is larger than 0, then I assume this invention fulfills the requirement of novelty and non-obviousness

producers are found, and/or whether the technology is understood fully depends positively on r. I denote the probability of finding a partner as well as any discount representing that the producer does not understand the technology fully, by $\pi(r)$ with $\partial\pi/\partial r > 0$.

Based on the strength of the (marginal) effects of Δ and r on V and π, this simple model makes straightforward predictions: universities as non-practicing entities will patent innovations with larger innovative steps, compared to firms using innovations in-house. This is due to the fact that universities need to find a licensee to achieve the value V while an in-house producer always derives the benefit V if the patent is granted. This finding is confirmed in the existing literature on patent citations Trajtenberg et al. [1997]. I also observe that an in-house producer will always choose the minimum level of disclosure, while a non-practicing entity interested in licensing the technology may choose a higher level of r, i.e. $r > r^*$, to increase the probability to find a licensee. Similarly, a non-practicing entity interested in generating income from holding-up firms that unintentionally infringe a patent, that is, a patent troll, will choose $r = r^*$. These considerations inform our research design/hypothesis as outlined below.

Hypothesis: Linguistic measures of patents applied for by universities are significantly different, indicating a better readability, than the comparable patents applied for by practicing entities.

5.3 Data

5.3.1 Sample

To conduct this analysis, I use full text patent documents, made available to me via The Lens (http://www.lens.org). The Lens is a joint service provided by QUT and the non-profit Cambia.org. Among other data set, The Lens covers all USPTO office patent applications since 2001, all patents granted from the USPTO (since 1976), from the European Patent Office since 1980, and the complete set of IP Australia, to name those relevant in our analysis. The data contains meta data, for example, name and organisation of an applicant, size of the patent family, etc. and can be cross-referenced with the patent

as well as the academic literature.

In order to control the industry diversity, all the patents I chose are from nanotechnology industry (class 977 of US Patent Classification). Nanotechnology is a relatively new research area, and both universities and corporations spend a lot of resources in this field, hence, I could gather enough patent samples. Besides, Ouellette [2012] did an impressive study to analyse whether patents are used as a source of technical information in nanotechnology industry. She asked a researcher in nanotechnology industry to read 4 patents, and check whether these patents contained enough technical information, and whether these patents are easy to read. Following her research, I would like to study the reason(s) that probably sit behind the difference of disclosure level. That's why I use nanotechnology industry as a starting point. I collect all the samples which were granted from 2009 to 2012 to control the yearly diversity to make up the data set. There are 3029 patents granted during this period and 1697 of them are domestic applications. I then exclude all the patents that applied by individuals, departments of government, research institutions, non-practising firms, and those the category of patent applicants are not sure. Finally, my data set is made up of 487 patents applied by universities and 1014 patents applied by practising firms. Given some universities and practicing firms have more than one patent in my dataset, and it is very possible that the readability level of patents from same patentees may not be independent, and residuals may also not be independent, I use the cluster option to indicate that all the patents are clustered into patent owners to make sure the observations would be independent between patent holders.

5.3.2 The Readability Measures

I run the computational linguistic program to the whole document of all the 500 patents to test whether the language used in patent documents is significantly different in terms of readability between two groups (universities vs practicing firms).

As we discussed in the section Background, the 106 features contained in the linguistic model have been used as the index of readability separately or jointly in previous research, and these features have their own focus. (These 106 features can be classified into four

groups: lexical richness and POS features; syntactic complexity features; disclosure features and word characteristics features) Since these features have not been used in patent documents before, one single feature may not be able to check the difference of readability accurately. Therefore, the basic rule of using these features is that I will examine the value of all these 106 features for each sample group (university patents versus practicing firms patents) and their differences, and see how many of them indicate that university (or practicing firms) patents are more readable (higher disclosure level).

Based on the values from linguistic model, I use t-test to test whether the mean of each feature is the same within the two groups (university patents vs practicing firm patents) and their differences. Here the main concern is the sample size is unequal between these two groups (487 vs 1501), and the Student's t-test may be unreliable, since variances differ between these two groups. This concern can be overcame by the use of unequal variance t-test [**?**].

Within the 106 features from the linguistic model, the traditional Fog index is probably the most popular feature. The Fog index was developed by Robert Gunning Associates, a consultancy founded by Robert Gunning in 1944. The formula of Fog index is as follows:

$$Fog = 0.4 * (words_per_sentence + 100 * percent_of_complex_words)$$

As we can see, the Fog index is a function of words per sentence and percentage of complex words (more than two syllables). This formula estimates the years of formal education that the reader needs to understand the documents on first reading. So the number of Fog can be explained as follows: if a document has a Fog index score of 6, then it means it should be easily readable by those who have 6 years of education. Usually, 12-14 means ideal; 14-18 means difficult; larger than 18 means unreadable.

Fog index is one of the most widely used linguistic features [Loughran and McDonald, 2014]. In the area of accounting and finance, the papers using Fog Index include Li [2008], Biddle et al. [2009], Miller [2010], Lehavy et al. [2011], Dougal et al. [2012]

and Lawrence [2013]. Biddle et al. [2009] even claim that Fog Index is "a measure of financial statement readability". Fog is the main readability feature I use in this thesis. Of course, I still use the other 105 features to consolidate my linguistic analysis results.

5.4 First Results

5.4.1 Are University Patents More Readable Than Those of Corporations?

Based on the values from linguistic model, I use t-test[4] to test whether the mean of each feature is the same within the two groups (university patents vs practicing firm patents) and their differences. Some may argue that technologies contained in practicing firms' patents are more complicated than those of universities, so practicing firms' patents are harder to understand correspondingly. To address this argument, I use a simple OLS regression model to test whether the difference within two groups can still hold when controlling the value of patents. Here I introduce family-size of the patent (the number of jurisdictions in which the patents are sought for protection) as the index of technical value of patents. Family-size, together with other indicators, such as the number of forward citations, renewal behaviour and so on have been used to approximate the patent valuation for a long time [Harhoff et al., 2003, Putnam, 1996, Trajtenberg, 1990]. The reason that I use family size here is, within the group (universities or practising firms), the disclosure levels that applicants would choose based on their assessment on patent value at the stage of drafting, that is the subjective value of patents. Comparing to forward citations, family size is more a reflection of subjective value.

$$y_{ij} = \alpha_{ij} + \beta_{1ij} * industry + \beta_{2ij} * family + e_{ij}$$

where independent variable y is the value of every feature; "*industry*" is a dummy variable, industry = 1, when it is a practicing firm patent and industry = 0, when it is an

[4]Given I have relatively large sample size (statistically:487 patents applied by universities and 1014 applied by practising firms), and most of the features are normal distributed, and the others are not so undistributed. Therefore, I use ttest to test whether these two groups have the same mean on each feature.

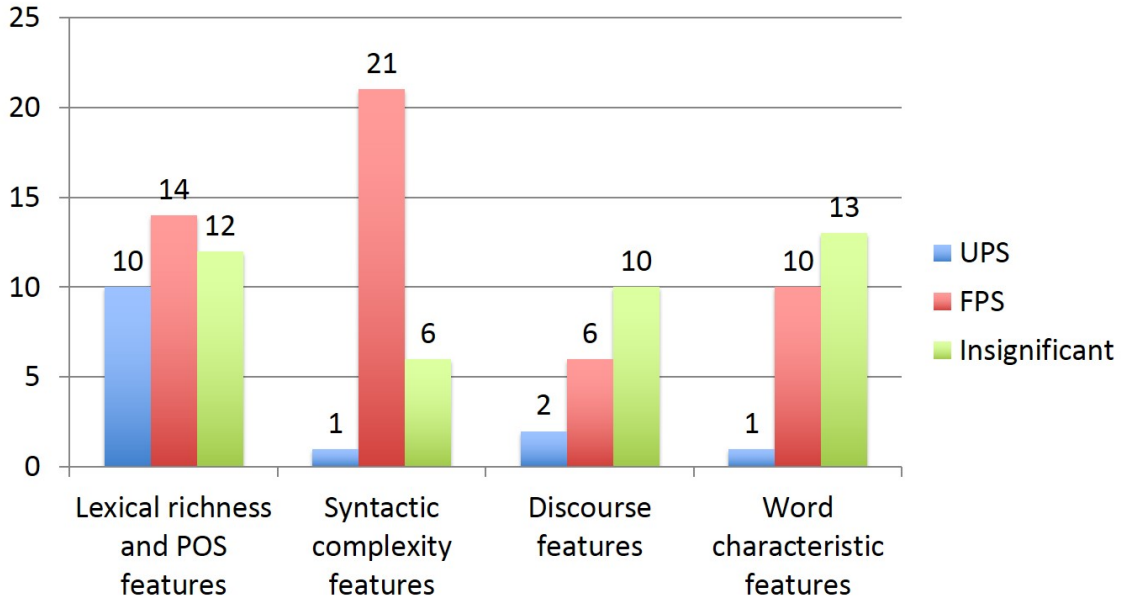

Figure 5.1: The linguistic analysis results of four groups

university patent; the "family" is the patent family size.

Among all 106 features, 65 features (61.3%) are significantly different from each other. Furthermore, among all these 65 features, 51 features (78.5%) suggest that practicing firm patents samples (FPS) are more difficult than university patents (UPS) (Figure 5.1)[5]. A full results table (which contains the mean of each sample, differences and percentage difference of mean values, and p-values from OLS regression) is provided in Appendix B.

Examining each group of features more carefully, we can see that in the group of lexical richness and POS features (including 36 features, and 24 features are significantly different), 14 features indicate that FPS have higher values, while 10 features support that UPS have higher values. For this group, the number of features which suggests FPS are more difficult is close to the number that suggests the opposite (14 vs 10), it is still relatively safe to argue that FPS are more likely to be more difficult than UPS.

In the second group (i.e., syntactic complexity features which indicate the level of

[5]For all the features, except TRAD_Flesch (the traditional Flesch formula), higher value means more difficult. Therefore, we shall expect a positive coefficient for "industry", if FPS are more difficult.

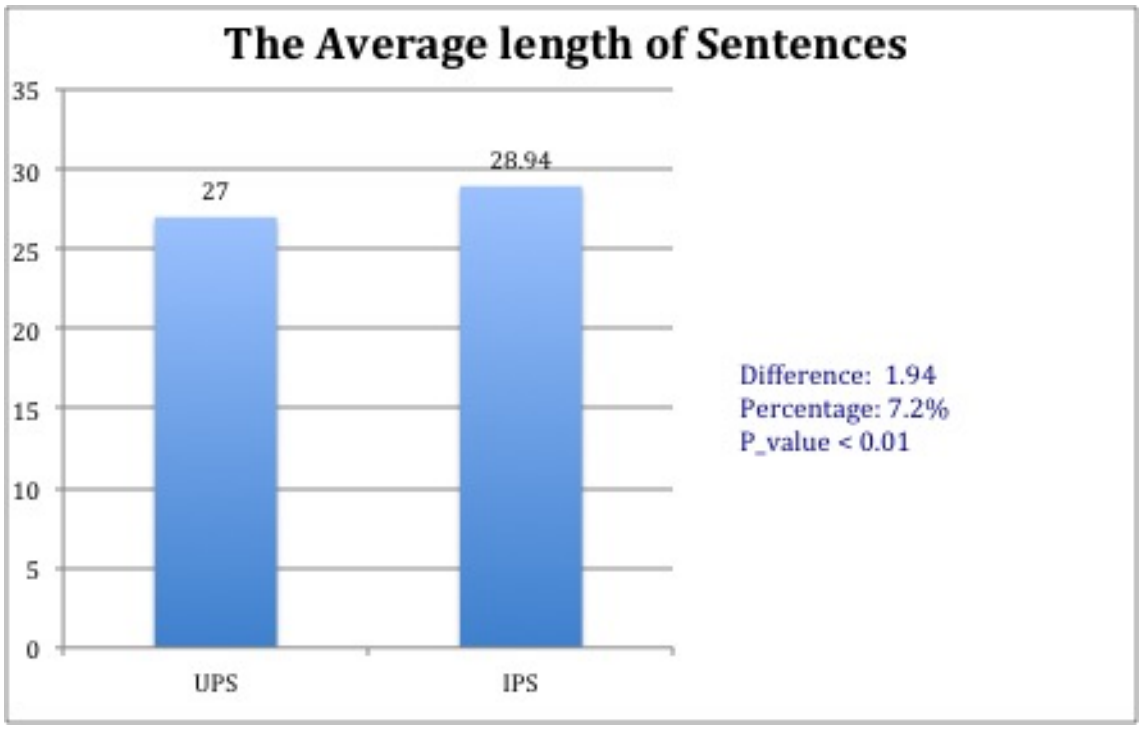

Figure 5.2: Average length of sentences

difficulty about the structure of sentences,there are 28 features in this group.). Twenty-two features have significant differences and 21 of them show that FPS have higher values. For example, as can be seen from Figure 5.2, the average length of sentences in UPS is 27 while it is 28.94 in FPS. FPS is 7.2% longer than UPS (p-value < 0.01). Another interesting example is the average height of a parse tree of the sentences (Figure 3). The average height is 11.44 in FPS, and is 4.9% higher than in UPS (10.91) (p-value < 0.01). Besides, the number of clauses per sentences, the number of wh-phrase of sentences, the number of clauses of t-units, the average length of noun phrases and the average length of t-units are all higher in FPS than UPS, which strongly support that the structure of sentences is more complicated in FPS.

For the group of discourse features (18 features), six of eight features which are significantly different support that FPS have higher values, including local argument overlap, local content word overlap, local noun overlap and local stem overlap.

In the group of word characteristics (24 features), 11 features have significant differences and 10 of them tell that FPS have higher values, including automated readability

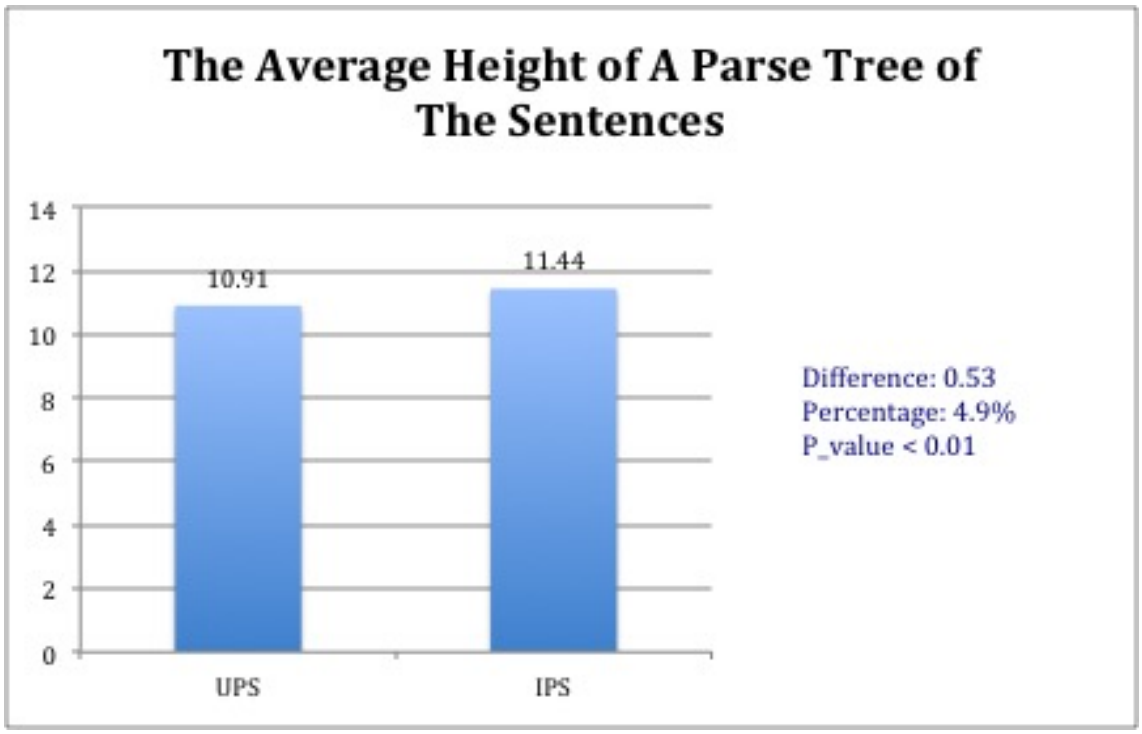

Figure 5.3: Average height of a parse tree of the sentences

index, Flesch index, Kincaid index, LIX index and so on. In these features, the other very famous formula-the Kincaid formula- is 11.24 for UPS while 11.89 for FPS, and FPS is 5.8% higher than UPS (p-value < 0.01).

In conclusion, the majority of features which are significantly different within two groups suggest that FPS are more difficult and this still holds in each group. Therefore this empirical evidence supports my hypothesis that FPS shall be more difficult.

5.4.2 Robustness Tests

To consolidate my result above, I double check whether this difference still exists in other industries. I then apply the computational linguistic program to patent samples which I collected for my third project. My dataset for the third project contains 1000 patent applications which applied to USPTO in the pharmaceutical industry in 2010. To make sure the randomization of samples, I pick all the 796 applications published in January 2010, and another 204 applications from February based on the patent publication number

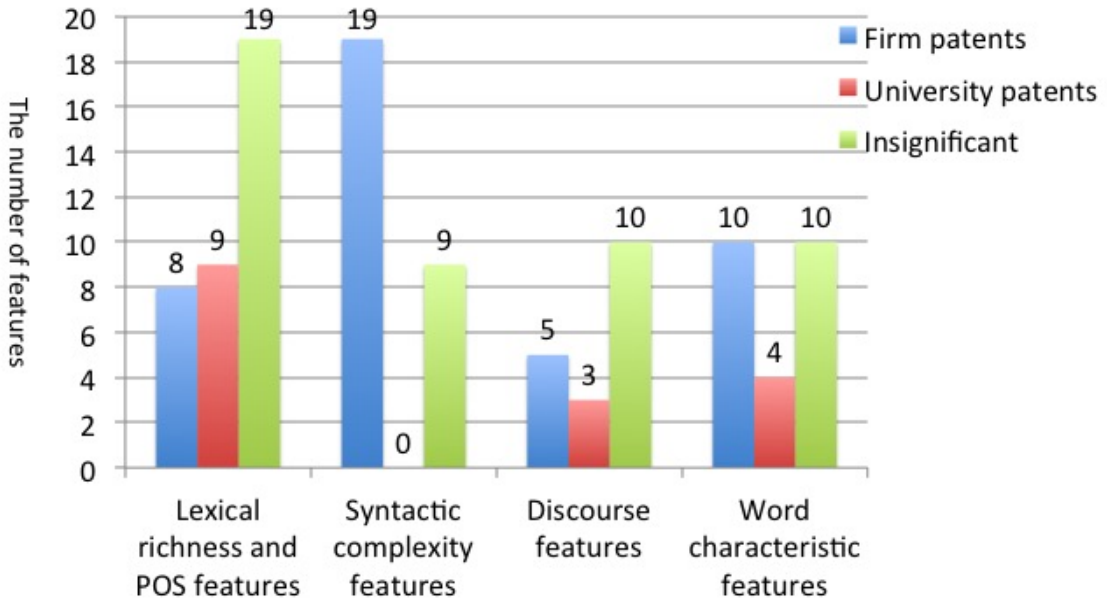

Figure 5.4: The linguistic analysis results of four groups

order. For the other details of the dataset, please see the Data section in Chapter 7. I choose all the American applications which applied for by universities and practicing firms and see whether the results above still hold in this new dataset. There are 86 patent applications applied for by universities and 277 from practicing firms.

I use the same method to test whether the mean of each feature is the same within two groups (university patents vs practicing firm patents) and their differences. And I use the same OLS model to test whether the results still hold when controlling the value of patents.

Among all 106 features, 59 features (56%) are significantly different from each other. Furthermore, among all these 59 features, 43 features (72.9%) suggest that practicing firm patents samples (FPS) are more difficult than university patents (UPS) (Figure 5.4). A full results table is provided in Appendix C.

Examining each group, we can see in the group of lexical richness and POS features (18 features out of 36 are significantly different between UPS and FPS), 8 features indicate that FPS are more difficult, while 10 features support that UPS are more difficult. For the group of syntactic complexity (19 features out of 28 are significantly different), and all of

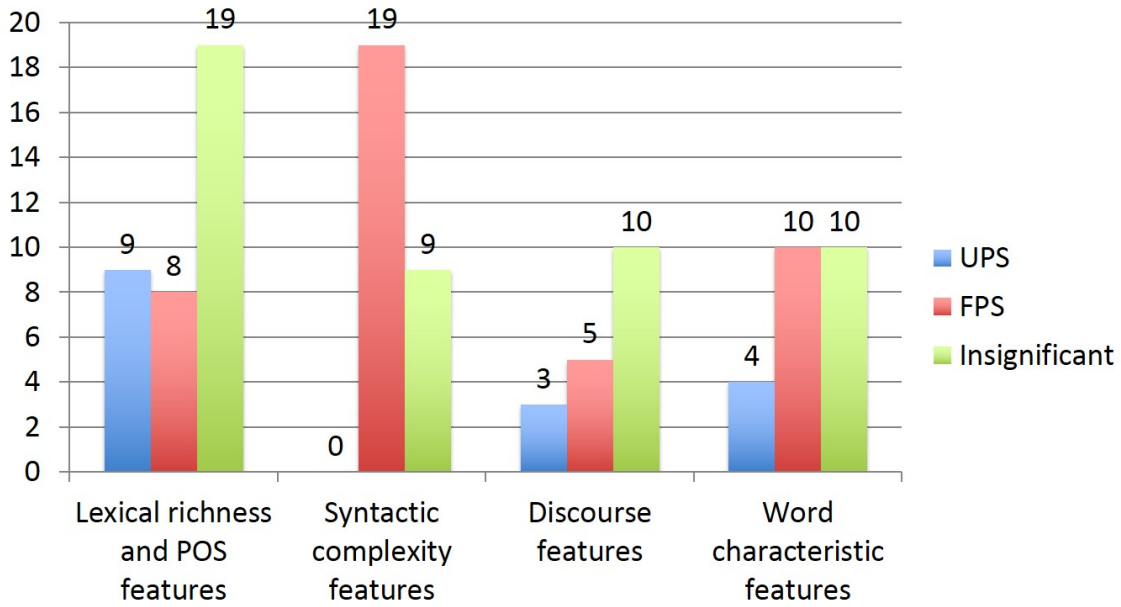

Figure 5.5: The linguistic analysis results of four groups

these 19 features support that FPS are more difficult. For the group of discourse features (8 features out of 18 are significantly different), and 5 features support FPS are more difficult. For the group of word characteristics (14 features out of 25 are significantly different), 11 features support FPS are more difficult[6].

Based on the results above, it is clear that the difference of disclosure level between UPS and FPS still exists in the new dataset, and this supports the robustness of my results.

5.4.3 Readability Features

Given this is the first time that computational linguistic analysis method has been introduced into the area of patent documents, it is helpful to check how the linguistic program works after running on two data-sets.

There are 11 features that don't work in both datasets. The value of these features are all zero, and there features are: DISC_RefExprPerProPerWord, DISC_RefExprPossProPerWord, DISC_RefExprPronounsPerWord, POS_numInterjections, POS_numPerPronouns, POS_numPronouns,

[6]Within these features, The mean of Fog index is 19.88 (median: 19.72) for practicing firm patents and 18.75 (median: 18.32) for university patents (p_value<0.02), which means practicing firm patents are less readable than university patents.

Table 5.1: Significant Features in Each Group

Group	No. of features	First Dataset	Second Dataset
Lexical Richness and POS Features	32	24(75%)	17(56.3%)
Syntactic Complexity Features	26	22(84.6%)	19(73.1%)
Word Characteristics	22	11(50%)	14(63.6%)
Discourse Features	15	8(53.3%)	8(53.3%)

POS_numWhPronouns, SYN_numConstituentsPerSen, SYN_numRRCsPerSen, WN_numHypernyms, WN_numHyponyms. 4 features are from the group of lexical richness and POS features, 3 from the group of discourse features, 2 from the group of syntactic complexity features, 2 from the group of word characteristics.

Apart from these features, there are 65 features (68.4%, out of 95 features) that are significantly different between two groups for the first dataset and 59 features (62.1%) for the second dataset. Examining each group, for the first dataset, the percentages of features that are significant are 75% for the group of lexical richness and POS features, 84.6% for the group of syntactic complexity features, 50% for the group of word characteristics, 53.3% for the group of discourse features. For the second dataset, the percentages are 56.3%, 73.1%, 63.6% and 53.3% respectively (See Table 5.1).

The percentages of features which are significantly different between two groups in two datasets and in each of four groups can support that the linguistic model works quite well in general.

Besides, within the features that are both significantly different between two groups (universities versus practising firms) in two datasets, 43 of them are the same features, this means 66.2% of the features that are significant for the first dataset are also significant for the second dataset. Among these 43 features, 42 features give us the same results (same sign) while only one feature gives the opposite result. This suggests that the linguistic model I use is sensible and stable.

If we go deep to examine each group, it is obvious that the group of syntactic complexity features works best in both datasets, the percentages of features that are significant in this group are the highest in both datasets. I then check the linguistic results in both

datasets, except the group of lexical richness and POS features[7], the other three groups, the groups of syntactic complexity features, word features and discourse features all give us a clear answer: significantly more features support that practising firm patents are more difficult, especially for the group of syntactic complexity features, 21 features (out of 22 features that are significant) in the first dataset, and all the 19 features that are significant in the second dataset support that practising firms are more difficult to read. This also supports that this group works best in both datasets.

Apart from the features in the group of syntactic complexity features, traditional readability formulae within the group of Word Characteristics also work pretty well in both datasets. There are 10 traditional linguistic features: ARI, Coleman, Fog, Forcast, Flesch, Kincaid, LIX, Smog, NumChars(the average number of characters per word), NumSyll(the average number of syllables per word) formula. The traditional readability formulae are all based on sentence length, word length, number of hard words (multi-syllables) or the combination of them[8]. Based on their mathematical formulae, these features usually can give us an output number to tell how many years of education a person needs to understand a piece of writing[9].

For the two datasets, there is no significant difference on the average number of characters per word, the average number of syllables per word is slightly higher for the patents applied for by practising firms (1.25%). This means for the average number of syllables and characters per word, there is not too much difference between university patents and practising firm patents. And this also suggests that the differences on the other traditional readability features are more from the difference on syntactic complexity.

[7] 10 features support university patents are more difficult while 14 features support practising firms are more difficult for the first dataset, and 10 features versus 8 features for the second dataset.

[8] In these features, the formulae of Forcast, NumChars, NumSyll and Smog are based on single measure: Forcast formula is based on the number of single-syllable words, NumChars formula is based on the average number of characters per word, NumSyll is based on the average number of syllables per word, Smog formula is based on polysyllable count. The other features are all the combination of two measures. The formulae of Flesch, Kincard and Fog are all based on average sentence length and number of hard words. The formulae of Lix, Coleman and ARI are both based on word length and sentence length.

[9] Some may concern that all the value of these traditional features will be bunched at the very top of the scale, and I will get very little statistical variation from these features. Actually if we take Fog index as an example, the mean of university patents is 18.04, with a range of 11.88 (from 12.94 to 24.82), and the standard deviation is 1.77. The result is similar for practicing firm patents. Given New York Times has an average Fog Index of 11-12, it seems Fog index still works well for patent documents.

Actually, this makes sense in reality. The readers of patents are other researchers or managers in the R&D department. The disclosure requirement in US patent law also require that patent specifications shall enable "any person skilled in the art" to understand the technology. Given readers of patents are all researchers and managers, and they are persons "skilled in the art", they should be able to understand those technical terms or other multi-syllables words, and the thing matters shall be the syntactic complexity.

In conclusion, the linguistic model works quite well in both data-sets. However, it still has its disadvantages, such as the possible overlapping problem of features, and whether all these features shall have the same weight, and whether we can get a single readability index based on all these 106 features, and I will keep improving this model with our linguistic experts.

5.5 Does Readability Affect Knowledge Flows?

After proving that different disclosure rates are chosen by different applicants based on their own incentives, a further question which can be asked is whether the different levels of disclosure will affect the knowledge flows, since this is the main purpose of patent disclosure. In their seminal paper, Jaffe et al. [1993] point out that patent citations can be used as a paper trail of knowledge flows. Therefore, forward citation has been widely used as a index of knowledge flows. Here in this thesis, I also use the number of forward citations as the index of knowledge flows and study the relationship between disclosure level and forward citations. My assumption is that there should be a positive relationship between patent disclosure level and knowledge flows.

Since the forward citations are count data, and is over-dispersed (the mean of forward citation is 5.54, while the standard deviation is 7.83, so obviously, the variance is much greater than the mean.), I use negative binomial regression model to examine the relationship between the number of forward citations and the level of difficulty and how the number of forward citations varies across different types of patent samples (industry vs university).

$$y_{ij} = \alpha_{ij} + \beta_{1ij} * readability + \beta_{2ij} * industry + \beta_{3ij} * year + e_{ij} \quad (1)$$

where y is number of forward citation which I use as the knowledge flow index; For readability, I use Fog index first[10]; "industry" is a dummy variable, industry = 1, when it is a industry patent and industry = 0, when it is a university patent; year is the patent granted year.

I further add a interaction term "industry*readability" in model 1 to check whether the effect of readability on knowledge flows (y) is different in different group (university vs industry).

$y_{ij} = \alpha_{ij} + \beta_{1ij} * readability + \beta_{2ij} * industry + \beta_{3ij} * readability * industry + \beta_{4ij} * year + e_{ij}$ (2)

However, the coefficients of readability in these two models are all insignificant, which means that we cannot find there is a relationship between the disclosure level and knowledge flows. I further replace the Fog index with every other feature in my linguistic model and double check the results. For the model 1, the coefficient of readability index is significant for only 32 features, and 18 of them support that lower disclosure level of patent documents leads to less forward citation, and 14 of them support the opposite relationship. If we remove the coefficients that are extremely small (equal or less than 0.01), the final result becomes 18 versus 8.

For the model 2, after adding the interaction term, the coefficient of readability index is significant for 26 features, and 9 of them support that lower disclosure level of patent documents leads to less forward citation, and 17 of them support the opposite relationship. Same as the model 1, if we remove the coefficients that are extremely small (equal or less than 0.01), the final result becomes 9 to 11. The coefficient of the interaction term is significant in only 9 features, and all of them are negative which indicates that disclosure level has a smaller effect to knowledge for industry patents.

[10]The reasons why I use Fog index here are: 1) as I discussed in the Literature review, Fog Index has been widely used in financial research; and 2) I need a simple but effective index as a measure to control for the overall level of readability of each patent document.

Given that there are not enough features that are significant (26 for model 1 and 20 for model 2), it is hard to make a clear conclusion that whether there is positive relationship between disclosure level and knowledge flows.

5.6 Conclusion

In this chapter, I study whether patent applicants would strategically choose the disclosure level of their patent applications at the stage of drafting. Based on their business models, my theoretical model suggests that universities have more incentive to better disclose their patents which can signal the technical value of their inventions than practicing firms. The empirical evidence from nanotechnology industry and pharmaceutical industry both support the finding that patents applied for by universities are significantly more readable than those of practicing firms. The results of this project tell us the difference of disclosure level between different applicants is not random and this gives us a relatively safe starting point for the further research of patent disclosure. Besides, the results from two datasets also support that the linguistic analysis model is effective and stable.

Of course, since this is the first time that computational linguistic analysis is being used for patent documents, together with the limited time and resource, there are still some limitations in this paper. First, our results might suffer from the imperfect matching of linguistic analysis model for patent documents. In general, it works pretty well, but there are still some problems: 1) the problem of overlapping features. There are 106 features in the model and some of them, like Fog formula, Kincaid formula and Flesch formula all based on the combination of sentence length and word length[11]. This will weaken the effectiveness of comparison results, since the results of group comparison (universities vs practising firms) are based on the number of features. As the next step of this research, this question needs to be addressed. 2) whether all these features shall have the same weight and whether I can get a single readability index based on this linguistic model (or which feature in this model can be used as the unique readability index which grasp the special characteristics of patent documents). Apart from group comparison, a

[11]The correlation between Fog index and Kincaid index is 0.97.

single readability index is needed when we try to study the effect of readability, just like how Fog index is used in finance and accounting. It is worth to further test which feature is best specified in patent documents.

The empirical evidence does not unveil the possible positive relationship between disclosure level and knowledge flows, because too few features are significantly different. The reason for this may be the use of pooled citations. In reality, a large part of forward citations is added by patent examiners. Alcacer and Gittelman [2006] point out that 63% of citations were actually added by examiners and 40% of all patents have all forward citations added by examiners, and they have already suggested that researchers might suffer from the bias brought from using pooled citations. I will test this again by using inventor citations alone as the next step.

Chapter 6

An Linguistic Analysis of Non-practicing Entities' Patent Acquisitions

6.1 Introduction

Non practicing entities (hereafter shortened to NPEs) are firms that acquire patents from other entities (especially from individual inventors or small firms) or from their own research (very few), but never or rarely practice their patents by themselves [Shrestha, 2010]. NPEs attempt to generate income from suing or threat to sue other practicing entities (hereafter shortened to PEs) for patent infringement [Cohen et al., 2014, Hagiu and Yoffie, 2013, Lemley and Shapiro, 2006, Reitzig et al., 2007, Watkins, 2013]. They are also known as "patent assertion entities" (PAEs) or "patent trolls"[1] [Yeh, 2012]. Many papers use patent trolls directly. Since the role of NPEs is still unclear and being argued over, I prefer to use NPEs: more neutral than using patent trolls.

NPEs are the main driver behind the soar of patent litigation recently in the US[2]. Based on RPX's report on NPE litigation (2015), NPEs have already filed more cases than PEs

[1]NPEs usually refer to patent assertion entities (PAEs), universities and research institutions, independent inventors and Non-competing entities (NCEs), and here, NPEs equal to PAEs.

[2]The most famous patent troll case is the case of NTP vs. RIM, NTP, a small Virginia based firm, sued RIM, the company who is the maker of famous Blackberry phone, for patent infringement in 2001. The patents in disputes had never been used by NTP before the lawsuit. Finally, after 5 years, RIM settled the litigation in 2006 by paying NTP 612.5 million USD to avoid the possible permanent injunction NTP might be granted.

in the last four years. In 2015, NPE cases had even accounted for more than two thirds of all patent infringement cases(3621/5219)[3]. Besides, NPEs got almost three times larger damage award than PEs[4].

The rise of NPEs has sparked a bitter public debate on its role and impact in innovation system, which makes NPEs the most controversial patent intermediaries nowadays [Hagiu and Yoffie, 2013]. Some suggest that NPEs help inventors, especially individual inventors to earn returns from their own research, because they normally lack resources to commercialize their technology or license them to big companies under arm-length contracts [Shrestha, 2010]. However, more frequently NPEs are alleged to be a threat to innovation [FTC, 2003, Lemley and Shapiro, 2006]. Among the main criticisms, the most prominent one is NPEs' notorious "nuisance value" litigation strategy, where they sue a lot of companies simultaneously for relatively low amounts of damage to induce defendants to settle the cases rather than to go to trial [Hagiu and Yoffie, 2013].

Till now, scholars have not formed a conclusion on NPEs' role. Some scholars try to study the role of NPEs based on empirical data. Risch [2012] studies the patents that are used by NPEs in patent litigation, and they find that those patents look very similar to patents used by other firms in other litigation. Similarly, Fischer and Henkel [2012] focus on NPEs' patent acquisitions to study whether NPEs do rely on low quality patents, and they find that actually patents acquired by NPEs have a higher technological quality than those patents acquired by practicing entities. On the contrary, Bessen and Meurer [2013] estimate the direct costs of NPEs litigation based on a survey and a database of litigation, and they find that US firms accrued 29 billion USD of direct cost from NPEs litigation. Cohen et al. [2014] also find that NPE litigation has a negative impact on innovation of targeted firms: firms substantially reduce their investment on innovation after they settled with NPEs. Although these empirical research do not lead to a consistent result, we can find that the main focus of these research are studying whether NPEs attempt to own, in particular, patents that are low in quality and try to using these low-quality patents to

[3]See:https://www.rpxcorp.com/wp-content/uploads/sites/2/2016/07/RPX-2015-Report-072616.FinalZ.pdf

[4]See https://www.pwc.com/us/en/forensic-services/publications/assets/2016-pwc-patent-litigation-study.pdf

get disproportional profits. By answering this question, we will be able to study whether NPEs are truly "trolls" and shed light on the further policy consideration.

However, when patent value is measured in these research, they always refer to the inventive quality, very few of them study the other dimensions of patent value. Actually, there are many definitions about the quality of patents. For patent attorneys, a high-quality patent may refer to a well-written patent with clear claims and specifications which describe its protection scope clearly and this can make sure the patent can withstand from the future challenge. For engineers, a high-quality patent may refer to a major invention patent rather than an incremental step patent. For lawyers, a high-quality patent should be a patent that can withstand in a future invalidate challenge. For economists, a high-quality patent may need to increase the total welfare of society. Here we can roughly put these definitions into two dimensions: the inventive quality and the disclosure quality. The inventive quality refers to the technology itself which is contained in the patent. Usually, the traditional patent value indicators, such as patent forward citations, patent family size, are all used to measure the inventive quality. The other is "disclosure" value: whether the patent document clearly describe the technological information in terms of readability to make sure that a person having ordinary skills in the art is able to apply it (from the perspective of better disclosing technological information), and whether the patent claims, together with specification as detailed explanation, clearly describe the boundary of the scope of the patent right (from the legal perspective).

Encouraging inventors to disclose their new inventions is one of the top two economic justifications of the patent system. Apart from disclosing technology contained in patent documents, patent disclosure is very important from the legal perspective. A patent, as an intangible property right, is much harder to define the details and boundary than tangible property, and this makes the disclosure of the patent documents essential for the outsiders to understand the technology contained in the patent and avoid unintentional infringement. Therefore, a patent can be found non-enabling by the court if it fails to meet the disclosure requirement. Disclosure value would matter when the patent is challenged in the future. However, most of the extant empirical study on NPEs focus on the first

value, and no research has ever touched on the latter one.

Therefore, I focus on NPEs' patent acquisition and study whether disclosure level (readability) is one of their considerations. Given the NPEs' "nuisance value" litigation business model, one would expect NPEs to own, in particular, patents that are obscure and difficult to read, as this could lead to larger explanation of patents and bring more suits base on that Watkins [2013]. Therefore, my assumption is that higher disclosure levels produce less ambiguity in the boundary of patent right, and NPEs would choose to acquire patents with lower disclosure levels based on their business model.

In this chapter, I provide the first evidence on the readability of patents which are acquired by NPEs, and test whether NPEs tend to acquire linguistically weak or unclear patents (low quality in term of "disclosure quality"). I begin with a simple theoretical model of a patents market in which independent inventors supply the patents and NPEs and PEs decide which patents they would like to acquire based on the technical and disclosure quality of patents, and their business model as well. Our theoretical model supports that it is true that NPEs would end to buy linguistically lower quality patents compared to PEs based on their different business models. I also test whether the empirical evidence would suggest this opportunistic behavior as well. I work with three independent sources of data about the NPEs' patent acquisition behavior: the list of NPEs from RPX Corporation reports, patent assignment history from USPTO Patent Assignment Database, and the full patent documents from Lens.org. I use computational linguistic analysis tools to the full documents of NPE patents as well as PE patents (as a control group) to quantitatively measure the readability of patent text, and see whether our theoretical hypothesis can be supported by empirical evidence. However, linguistic analysis results do not support that patents acquired by NPEs are significantly poorly written than those acquired by PEs.

6.2 Background

6.2.1 Behind the Rise of NPEs

There are many reasons for the rise of NPEs. The key advantage of NPEs is they are immune to counter-assertion claims. For PEs, when they try to sue other PEs through patent infringement litigation, they need to bear the risk of counter-assertion claims, since the other PEs may have their own patent portfolios, and they can retaliate with patent infringement litigation as well and ask for the same remedy, such as injunction [5]. The so-called MAD (mutual assured destruction) strategy would make PEs think twice when they want to approach to the court [FTC, 2003]. On the contrary, NPEs do not make or sell anything, their core business is patent enforcement. Unlike PEs, they have nothing to lose in a patent infringement litigation, and they can sue or threat to sue other PEs without any scruple.

The rise of NPEs can also be attributed to the very risky, costly and time-wasting patent litigation process. From 2011 to 2015, the average amount of time that parties need to wait for the judgement in patent litigation is 2.5 years[6]. The average patent litigation cost is above 1 million USD[7]. Even NPEs only win 8% of the merits judgments, and settle the majority of the cases [Allison et al., 2010], the uncertainty and high litigation costs will make defendants find that settlement is the best way to get rid of the suit.

The third reason behind the rise of NPEs is the difficulty of defining patent property rights. As I discussed above, it is hard to define the boundaries of a patent as an intangible property right, and this is the so-called "notice failure" problem [Yeh, 2012]. The notice failure, together with rocketed patent granted number make it almost impossible for firms to do the patent search thoroughly to avoid infringement, and this gives NPEs huge room to enforce their patents.

[5] We see this happening a lot in reality, especially in the smart phone industry, for example, the Apple Inc. v. Samsung Electronics Co. case.

[6] See https://www.pwc.com/us/en/forensic-services/publications/assets/2016-pwc-patent-litigation-study.pdf

[7] See http://www.wipo.int/export/sites/www/wipo_magazine/en/pdf/2010/wipo_pub_121_2010_01.pdf

6.2.2 The NPEs Controversy

Opponents of NPEs criticize NPEs as "patent trolls" [8]. They claim that NPEs use low quality patents to engage in "nuisance value" litigation [Chuang, 2006, Hagiu and Yoffie, 2013]: NPEs sue or threaten to sue many companies simultaneously for a relatively moderate damages to induce defendants to settle rather than to go to trial, because many firms would rather settle than engage in the long, expensive and risky litigation [Chuang, 2006]. The patent litigation is very expensive in the US where the legal fees can reach 500,000 USD for summary judgment and 4,000,000 USD or more for trial, not to mention the loss that a possible injunction or delay of going to market may bring to firms [Risch, 2012].

Second, NPEs attempt to ambush practicing firms to extract disproportional profits. Patent ambush used to refer to the behavior that a patent holder does not disclose their patent(s) to standard organization until the standard has been fixed. These patents are called standard-essential patents Lerner and Tirole [2015]. Here the ambush strategy of NPEs refers to NPEs intentionally choosing the best time to sue their targeting practicing firms, for example, just after investments have been made or locked in, or before the release of a new product [Geradin et al., 2011, Hagiu and Yoffie, 2013].

Third, NPEs exploit independent inventors in the situation of information asymmetry. Compared to independent inventors, NPEs are more professional on differentiating valuable or trivial patents, and they can purchase the patents with a lower price and use them to extract much higher profits [McFeely, 2008]. Actually it is true in the real world, Bessen and Meurer [2013] find that the low rate the funds NPEs get is transferred to outside inventors. Similarly, Cohen et al. [2014] find their empirical result does not support any significant benefit transfer that NPEs pass-through to end innovators. Therefore, independent inventors actually do not benefit from NPEs' action and all the benefit NPEs got would transfer to customers as increased costs and harm innovation as a whole [Shrestha,

[8] There is an interesting story about the name of "patent trolls". In 2001, Intel's Assistant General Counsel, an in-house lawyer, Perter Detkin, called Niro, a patent attorney, a patent troll. This is where the term "patent trolls" is original from. However ironically, Detkin became the co-founder of Intellectual Ventures, the largest NPE all over the world after that.

2010, Yeh, 2012].

On the other hand, the defenders of NPEs argue that NPEs, as intermediaries, can add liquidity to patent markets. In the market of patents, buyers and sellers have difficulty finding each other. Actually, under the pressure of getting financial returns from their academic research, universities have already started to license their patents to NPEs Ledford [2013]. And NPEs can mitigate the inefficiency and illiquidity of patent markets [Hagiu and Yoffie, 2013, McDonough, 2006]. They can help small firms and individual patent holders to get profit from their intellectual property. It is hard for small firms and individual patent holders to make arm-length contracts with large firms because of the imbalance of financial resources. Large firms would rather to infringe the patent directly than getting a license from small firms and individuals: small firms will not survive long under the high litigation cost [Golden, 2006, Haber and Werfel, 2016, McDonough, 2006, Shrestha, 2010].

Based on these disputes, policymakers including Congress and USPTO are both considering new regulations on NPEs. However, before any policy be introduced, further research is still needed to address the insufficient empirical research on the amount and impact of NPEs.

6.3 Literature Review

There is some interesting theoretical research focus on the business model of NPEs (patent trolls). Henkel and Reitzig [2010] provide a theoretical analysis of NPEs (patent trolls) business models. They use a game-theoretic model to study the interplay between NPEs (patent trolls) and manufacturers, and it shows that policy changes (stop legal loopholes and improve the efficiency in court practice) might be able to mitigate the problem that NPEs generate profits from low technological and legal quality patents, but NPEs can still keep their business model as long as they move to rely on patents with high technological and legal quality, that is using patents that are sophisticated enough in technologically

crowded industries. In another research, Turner [2011] uses an equilibrium model of invention, patenting and infringement in the context of monopolistic competition. With the problem of patent thicket[9], if one of the patentees is a patent troll, the rate of invention falls further. Therefore, patent trolls hurt not only manufacturers but also welfare generally as a whole.

The extant empirical studies on NPEs mainly focus on NPEs' patenting and litigation behavior and these research did not lead to a consistent result [Bessen and Meurer, 2013, Bessen et al., 2011, Cohen et al., 2014, Fischer and Henkel, 2012, Lemley and Shapiro, 2006, Magliocca, 2007]. Bessen and Meurer [2013] use data from survey results and NPE litigation database, and find that the direct cost from NPE patent assertions reaches 29 billion USD in 2011, and most of the defendants are small and medium-sized firms. Moreover, they find that the payment which small inventors received from NPEs only account for 3% of all revenues that NPEs get from defendants, which means small inventors actually benefit little from NPEs' patent assertions. Their findings are supported by Cohen et al. [2014]. In their research, Cohen et al. compare the changes of R&D of firms who were targeted by NPEs before and after litigation to study how the NPE litigation affects the R&D expenditure. They find that firms substantially reduce their R&D investment after settling with NPEs (or losing to them in court).

For the quality of patents that are acquired by NPEs, Lerner [2006] uses data of business method patents and find that patent holders are "exploiting the system to obtain and litigate financial patents of questionable quality ". In contrast to Lerner's results, Risch [2012] studies the patents used in 971 patent cases filed by the ten most active NPEs. He finds that contrary to expectations, these patents are not mainly from business models, but related to different technologies, and these patents appear to be of "equal or higher quality" than other litigated patents. Besides, the validity rates of NPE patents affirmed by courts are close to the rates of other patents. Similarly, Shrestha [2010] uses patent data from USPTO and Stanford Law School's Intellectual Property Litigation Clearinghouse

[9]Patent thicket is the result of cumulative inventions, it means many patents cover on the single technology, and this makes the firms who want to commercialize this technology should obtain licenses from multiple patentees.

(IPLC), and finds that litigated NPE patents are significantly more valuable than the patents litigated by PEs that share the same technological class. Fischer and Henkel [2012] also find similar result in their research. They compare 392 patents acquired by NPEs from 1997 to 2006 to 784 patents acquired by practicing firms, and find that NPE-acquired patents are significantly more valuable than those acquired by PEs.

As discussed above, extant research has not formed a conclusion on the role of NPEs. In this chapter, I further study the quality of patents acquired by NPEs in the perspective of disclosure level (readability), and check whether NPEs do rely on low-quality patents.

6.4 Model

Since my goal is to analyze whether comparing to PEs, NPEs would like to acquire patents that are low in disclosure quality, I start with a parsimonious model of a patent acquisition market in which independent inventors are the only supplier of patents and NPEs and PEs are the buyers. Big firms are excluded in this market since they can be both patent suppliers and buyers based on their strong in-house research ability and powerful commercialization skills. In this model, I focus on a very simple scenario, in which NPEs and PEs need to decide which kind of patents they would like to buy based on their business models. In this market, NPEs and PEs serve different business models. For PEs, they focus on commercializing the technology contained in the patent, and protect its intellectual property right in the future (to make sure its IP right will withstand in a potential challenge in the court). For NPEs, they try to gain profits by using "nuisance value" litigation, when NPEs sue or threaten to sue many companies simultaneously to settle rather than to go to trial, as I discussed above.

The patents' value is measured by two dimensions: T, the technical value of a patent, and L, the disclosure value of a patent. Technical value refers to the saved cost that the new technology contained in the patent can bring to the patent holder [10], and linguistic

[10] I assume that patents in this market are all process patents. Of course the saved costs assumption holds in product patents as well, the new products either reduce the costs or increase the quality with same or lower cost compared to the existing products.

value refers to the certainty that the clear description would bring when the patent is challenged in the future. Here I use readability as the index of disclosure value, since the more easily patent documents can be understood, the more likely patents have a clearer boundary of right.

NPEs and PEs need to decide their WTP (willingness to pay) (W_N, and W_P) based on the technical value (T) and disclosure value (L), according to their business model when they try to buy a patent

$W_i = f_i(t, l), i = N, P.$

For a PE, it would focus on both technical value and linguistic value. Based on its business model, PE will choose patents with higher technology (more producing costs to be saved), and after investment made on this technology, the other thing it must be concerned with is whether the patent right could withstand a future challenge. Therefore, PEs would like to acquire patents that are both high in technical value (T) and linguistic value (L). I assume

$\partial W_P/\partial t >0$ that is, the PE's willingness to pay increases with the size of inventive step T (the cost the technology can save), and

$\partial W_P/\partial l >0$ that is, the PE's willingness to pay increases with the disclosure value of this patent L (higher certainty to withstand in the future challenge).

On the other side, for an NPE, based on its business model, its "core business" is the "nuisance value" litigation, and it targets to settle the case rather than to go to trial. As discussed in section Background, according to US Patent Law, a patent shall be presumed valid, and the "burden of establishing invalidity of a patent or any claim shall rest on the party who asserting such invalidity"[11]. This means the burden of proof is on defendants, if they want to claim the patents are invalid in a patent infringement case. Combined with the long length and high cost of patent litigation, defendants actually have already been put at a disadvantage when they face an NPE plaintiff. The uncertainty of patent litigation also make it extremely difficult to predict the final trial result for defendants.

[11] See 35 U.S. Code §282

[Bessen et al., 2011] also support this uneven bargaining power between NPEs and PEs, and claims that the NPEs' business model relies more on "notice failure" than the patent quality [Yeh, 2012]. Based on the reasons above, it is safe to assume that technological quality is not the main consideration for NPEs. Empirical evidence can also support this assumption. Allison et al. [2010] find that when the patent litigation go to judgment on merits[12], the win rate of NPEs is only 8%, while other entities is 40%. This can imply that patents used by NPEs in patent litigation may not be as high quality as those used by other entities. For the disclosure (linguistic) quality, as I discussed above, if patent documents are obscure and difficult to read, it would cause confusion and lead to more unintended infringement [Bessen et al., 2011]. Therefore, I assume NPEs would prefer low-disclosure-quality patents. Above all, I assume

$\partial_{W_N}/\partial_l$ <0 that is, NPE's willingness to pay reduces with the disclosure value of this patent L, and

$\partial W_P/\partial t$ >0 that is, NPE may also want to acquire patents with high technological quality, but not as keen as PEs.

And I also assume there are two cut-off values (L*, T*) for both PE and NPE.

Based on the frame of this model, we can move to predict the patent acquisitions for NPEs and PEs (see Figure 6.1). Clearly, NPEs and PEs will choose to buy patents in area of 3 and 2 respectively: PE will acquire patents with higher technical value and higher linguistic value as well (area 2), and they would not bother to touch patents in area 3 where NPEs would be happy to take over. For areas 1 and 4, it is a bit complicated. Patents with higher linguistic value but lower technical value (area 1) are more likely to be acquired by PEs, because clear description is against NPEs' interests based on their business model while PEs may consider to buy them with lower price even their technical value is lower than the cut-off value T*. This goes against the model assumption above. However, it is not uncommon that PEs acquire patents only for a defensive purpose [Bessen, 2003]. For those patents with high technical value and lower linguistic value (area 4), NPEs are obviously interested in them. But we may not be able to fully exclude PE in this area,

[12]The final judgment based on the ultimate facts or state of facts

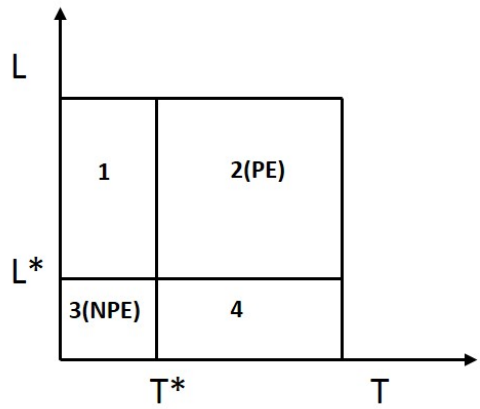

Figure 6.1: Model on patent acquisitions 1

since the high technological value might attract them to bear the risk of low disclosure quality.

Above all, we see that: PEs would acquire patents located in area 1 and area 2, and NPEs would end to acquire patents in area 3, and we may observe both PEs and NPEs in area 4. But overall, it is clear that patents acquired by PEs have higher linguistic quality than those acquired by NPEs. (See Figure 6.2).

6.5 Empirical Evidence

The goal of this chapter is to study the NPE patent acquisitions in the real world. To do so, I create a dataset to study the disclosure quality of patents acquired by NPEs.

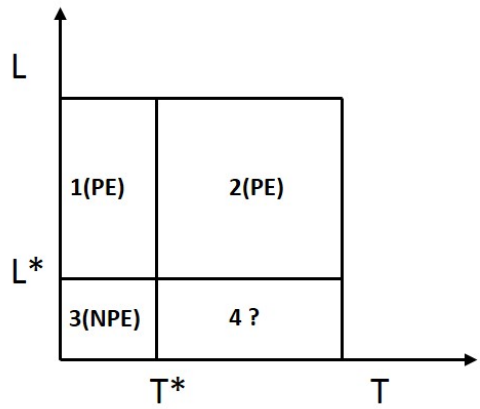

Figure 6.2: Model on patent acquisitions 2

6.5.1 Data

To test whether NPEs do rely on low-quality patents, I create a dataset of patents acquired by NPEs and PEs respectively. The dataset is collected from three different sources:

(i) RPX research report;

(ii) USPTO Assignment Dataset;

(iii) The Lens database.

The first thing to build up the dataset is to identify NPEs. I obtain information on NPEs from reports of the RPX corporation. RPX corporation is a for-profit firm, first publicly traded in May 2011. RPX has collected data about NPE behavior. The database of RPX is becoming more and more popular for patent research. I identify the names of NPEs from RPX reports. In order to make sure that all the NPEs in my list are active in their business, all NPEs are from the top ten plaintiffs (by campaigns led and total

Table 6.1: NPEs in our dataset

NPEs	No. of patents
EMPIRE IP LLC	22
POSITRON ACCESS SOLUTIONS INC	1
ACACIA RESEARCH GROUP LLC	23
BLACKBIRD TECH	17
CRYPTOPEAK	1
EDEKKA	1
EOLAS TECHNOLOGIES INC	1
HAWK TECHNOLOGY	1
MARATHON PATENT	1
MEDICI PORTFOLIO ACQUISITION LLC	5
MONUMENT IP BANK	2
OLIVISTAR	1
PPSITRON INTELLECTUAL PROPERTY	3
POSITRON TELECOMMUNICATION SYSTEMS	6
SELECT RETRIEVAL, LLC	1
VISUAL MENORY	1
WI-LAN/LABS INC	13

campaign defendants added) in 2015 and also in the previous 5 years as well [13]. There are 17 firms in total (Please see the Table 1).

Using USPTO patent Assignment Dataset, I collect all the patents that were acquired by these 17 NPEs during 2010 to 2015. In order to make sure that I can identify the incentive behind all these acquisitions, I remove all the patents that were acquired by PEs as well in their assignation history. The reason for that is because many patents had been acquired many times and assignees may include both NPEs and PEs, and this will bring noisy information when studying the incentive behind the patent acquisitions. Then for the second step, I match every NPEs patent with a patent that was acquired by a PE to form the other group. As with the group of NPE patents, the patents used in this group are only acquired by PEs in their assignation history. Third, in order to control the yearly diversity, every patent that is acquired by a PE in similar time ($\pm$ 2 years) with its matched NPE patent to make sure that the decision of acquisition was made under relatively the same law and policy background. And every NPE patent and PE patent was published

[13]Please see https://www.rpxcorp.com/wp-content/uploads/sites/2/2016/01/RPX-2015-NPE-Activity-Highlights-FinalZ.pdf, and https://www.rpxcorp.com/wp-content/uploads/sites/2/2016/07/RPX-2015-Report-072616.FinalZ.pdf

in the same year. Forth, I make sure that every NPE patent and PE patent is from the same US Patent Classification to control the industry diversity. These patents are from 40 different patent classes, and the patent classes vary from surgery to information security. Due to the strict control, I only get 100 patents each for the PE patents group and NPE patents group.

Then, I get the whole patent document from Lens (https://www.lens.org/lens/), and I follow the parsing procedure of Loughran and McDonald [2011] to remove all tables contained in patent documents. As with the first project, I run the computational linguistic program to the whole document of all these 200 patents to test whether linguistic measures of patents owned by NPEs are less readable than those of PEs.

6.5.2 Results

As with our first project, I examine the mean value of all 106 measures for each sample group (NPE patents versus PE patents) and their differences. I then use a simple OLS regression model to test whether these two groups are significantly different from each other on each of these features.

$$y_{ij} = \alpha_{ij} + \beta_{ij} * NPE + e_{ij}$$

where I use the value of every measurement as dependent variable and "NPE" is a dummy variable, NPE = 1, when it is a NPE patent and NPE = 0, when it is a PE patent. Since I have samples that are acquired by the same NPE or PE, and the values of features of these patents might not be independent, this could lead to residuals that are not independent as well. Therefore, I cluster all the patent samples into assignees.

Among all these 106 features, 11 are significantly different and 4 support that NPE patents are more difficult while the other 7 support that PE patents are harder to read (The Fog index is not among indices that are significantly different). The detailed results are present in the Appendix D. Although more features support that PE patents are harder to read (7 vs 4), we cannot conclude this just based on 11 features (out of 106 features). Therefore, the empirical evidence here does not support that NPEs may

particularly choose to acquire patents that are linguistically difficult to understand.

This may be due to the unbalance of NPEs patent samples. In the dataset, 75 patents are from 4 NPEs, out of 17 NPEs in total (EMPIRE IP LLC, ACACIA RESEARCH GROUP LLC, BLACKBIRD TECH, WI-LAN/LABS INC), and the number of patents acquired by these 4 NPEs varies from 13 to 23. The rest of the 25 patents are from another 13 NPEs[14]. This means the average disclosure level of NPEs patents group may mainly be decided by these patents. Therefore, I use OLS regression model again to test whether the mean value is significantly different between patent samples from these 4 NPEs and their control patents[15]. Eighteen features are significantly different, 11 features support NPE patents are harder to read, and 7 features support PE patents are harder to read. It seems for these 5 large NPEs, more features support their patents are harder to read, which supports the theoretical hypothesis. However, this result still suffers from the small number of features that are significant.

I further test whether NPE patents are more difficult than PE patents for each of these 4 NPEs. A simple OLS regression is used to compare the disclosure level. For Acacia Research Group, 12 features are significantly different, and 9 of them support the NPE patents are more difficult. For Empire IP, 17 features are significant and only 1 of them support NPE patents are more difficult. For Blackbird Tech, 8 features are significant and 6 of them support NPE patents are more difficult. For Wi-lan, 31 features are significant and 16 of them support NPE patents are more difficult. It seems for Acacia Research Group and Blackbird Tech, the patents acquired by them are more difficult to read than in the control group, and patents acquired by Empire IP are less difficult than the control group. However, the results still suffer from the small number of features that are significant, and we cannot have a clear conclusion just based on these features that are

[14]The reason of this unbalance is because of my strict control when I collect the samples. As I discussed above, I first pick the patents acquired by these 17 NPEs and only acquired by NPEs in their assignation history, and then match every NPE patent with a patent that is from the same industry and published in relatively the same year, and this control patent must be acquired by PEs only in its assignation history. If I cannot find a control patent for the NPE patent based on USPTO Assignment Dataset, I just remove the NPE patent in the list, and move to the next NPE patent. This is the reason that why I only get 100 patents for NPEs and PEs each.

[15]Given features are still normal distributed or are not very non-normal distributed, I can still use OLS regression model here.

significant.

I also run the linguistic model on patent specifications only, and the result is still the same. I could not find any significant difference between NPE patents and PE patents.

6.6 Conclusion

In this chapter, I study whether NPEs would like to acquire patents that are linguistically hard to understand based on their "nuisance value" litigation business model, and the theoretical model predicts that NPEs do end up to acquiring patents that are low in linguistic quality. However, I could not find there are significant differences on readability between patents acquired by PEs and NPEs in empirical results. This may be due to that NPEs are not able to discern the disclosure level of patents, and just buy every patent they can find on the market, or they acquire patents based on their technological value. And this may also due to the small sample size (100 each for NPE patents and PE patents). As the next step, I will test the readability level on a larger size sample, and see whether a clearer answer can be found.

Chapter 7

Does There Exist a Discrimination to International Patent Applications?

7.1 Introduction

National treatment is one of the fundamental principles in the international intellectual property protection system [Loewenheim, 2009]. For the patent system, it means that foreign patent applicants should not receive worse treatment than their domestic counterparts in any country [Scotchmer, 2004b]. The economic rationale for intellectual property is that it provides the innovator with a temporary monopoly to allow it to recoup its investment, and this will provide firms with greater incentive to invest in innovation, and to create a virtuous cycle. However, encouraging innovation is not sufficient for economic progress, new technologies need to be diffused [Tirole, 1988]. Patent property right has always been criticized as blocking the diffusion of innovation, since it gives the inventor a temporary monopoly position. And monopoly is recognized as a necessary evil, as it would inevitably cause deadweight loss (unless perfect price-discrimination is possible). Therefore, the patent system offers a trade-off between the marginal dynamic benefit with the marginal static efficiency loss, which requires patent applicants to disclose the patented invention in details Nordhaus [1969].

However, in an open economy, this trade-off changes. Patents are territorial rights

(WIPO), and every country can only grant protection within its own territory[1]. But overseas innovators also have the access to these technology (disclosure of patent documents) and the benefits of innovation would spread to other countries. This means that a country cannot reap all of the benefits from its domestic innovations, but bear all the marginal static efficiency loss. Therefore, every country would like to get the strongest possible protections in a foreign country, and offer weakest possible protections for foreigners in its own territory [Scotchmer, 2004a]. And this would obviously affect the global innovation and may also trigger the reciprocal action by other countries [Webster et al., 2014]. In order to solve or mitigate this problem, national treatment is used to make sure that equal treatment will be given to both domestic inventors and foreign inventors. In fact, national treatment has been established in almost all the most important conventions, including Paris Convention, the Berne Convention, the Rome Convention, the Universal Copyright Convention, TRIPS, NAFTA (The North American Free Trade Agreement), the WPPT (WIPO Performances and Phonograms Treaty), and in the WCT (WIPO Copyright Treaty) Loewenheim [2009].

Despite national treatment is so important, there are few economic research on it, and when studying international intellectual property protection, national treatment always be assumed to be given [Grossman and Lai, 2004]. There are some empirical research studying whether national treatment has been applied by certain counties. In their most recent empirical study, Webster et al. [2014] compare the examination outcomes at the EPO and JPO (Japanese Patent Office) on same patent applications, and find that domestic patent applicants are significantly more successful than foreign applicants in general, and especially in areas which have high technological specialization in the domestic economy. The same result is found by Liegsalz and Wagner [2013]. They find that domestic applicants achieved shorter patent granting time than foreign applicant, after controlling for other determinants at the State Intellectual Property Office of China. These results are troubling and surprising because all countries (as convention members) shall

[1]The U.S. patents can only be protected in the U.S., and if U.S. firms want to receive European patent protection, they must file a patent application in EPO within one year of the U.S. application. It is the same to ask for protection in Japan, China and so on. And it is same for foreign applicants if they want to ask for patent protection in the U.S.

subscribe to the principle of national treatment of all applicants. The source or sources of this domestic advantage are not understood, but one possible explanation is that foreign applicants are not as facile with the patent language, or that patent documents translated from another language are not as clear.

To disentangle this argument, I introduce the quantitative measure of readability into patent application data. More precisely, I collect 1000 patent applications (including domestic and foreign applications) from USPTO which were published in pharmaceutical industry (US Patent Classification Number: 424) in January and February 2010, and examine the readability of all these applications. I first compare the mean readability score for those patents with a domestic inventor and those that do not have a domestic inventor, and next run a regression analysis (probit) of the probability that a patent is granted namely controlling for readability, testing the hypothesis that there is no difference in the probability a patent is granted between applications with a domestic inventor and those without. I find that domestic applications do have a higher granted rate than international applications in our dataset, and in general, domestic applications are more readable than international applications. However, the domestic inventor advantage dose not exist after controlling the readability of applications. Although the result of this chapter still needs to be further consolidated, it provides a new perspective to study the domestic inventor advantage.

7.2 Background and Literature Review

National treatment is firstly well established in the Paris Convention of 1883 on patents and other industrial property [2]. Article 2 of Paris Convention requires that nationals from any member shall be treated no less than nationals in a certain member country, they should have the same protection and same legal remedy against any infringement of their rights. After that, almost all the most important conventions, including the Berne Convention, the Rome Convention, the Universal Copyright Convention, the TRIPS,

[2]The full name of Paris Convention: Paris Convention for the Protection of Industrial Property, and the full treaty can be seen from: http://www.wipo.int/wipolex/en/treaties/text.jsp?file_id=287556

NAFTA, the WPPT, and the WCT, all have national treatment as one of the fundamental principles. The TRIPS makes a further step on national treatment, it introduces the enforcement provisions, so the disputes about national treatment that can be brought before the WTO.

The rational of international intellectual property protection is to stimulate global innovation. As discussed before, because of the principle of territoriality of patent right, and the benefits of innovation that would spread across the boundaries of countries, countries would have an incentive to free-ride on other countries' R&D. Grossman and Lai [2004] have already pointed out that a country that has a very low share of the world economy has no incentive to grant patents, and the more independent countries within a international agreement, the more severe the free-riding problem would be. That is why we need protection of intellectual property at an international level.

From the point of law, since the harmonization of national patent laws is unrealistic, national treatment is a "simple" and "ingenious" solution for international protection of intellectual property [Loewenheim, 2009]. Because the national treatment just requires that foreign inventors should get the same protection as national inventors, it does not require that every country should provide equal protection, since different countries may have different patent systems and they may offer different levels of protection.

Despite national treatment being so important, there are few economic studies on national treatment. For theoretical studies, Aoki and Prusa [1993] find that offering "discriminatory" protection to domestic firms and foreign firms would not necessarily lead to greater R&D by domestic firms. Its effect on R&D is affected by many other factors, including whether firms currently have products on the market, the costs of R&D, the potential innovation's value and how efficiency a firm's R&D can be. After that, Scotchmer [2004b] studies the incentive of countries to offer national treatment to foreign inventors, and finds that countries, especially small countries, would not grant national treatment to foreign inventors without reciprocity, but a small country may be more receptive if national treatment is reciprocal. For the way to avoid national treatment, Regibeau and Rockett [2006] compare the effect of tariff and administrative delay (separately and

combined) on the timing of introducing new products into a market and the changes in welfare. Within their two-country two-firm market model, Regibeau and Rockett find that tariffs are better instruments to delay the foreign products entering into a domestic market than administrative delay. But in the context of trade liberalization, import tariff may be prohibited and discriminatory administrative may become the better choice for home country.

For the empirical studies, Yang [2008] compares the invention patent applications in China and U.S. on granting practise. She finds that both China and U.S. try to equally protect domestic and foreign applicants, but domestic applications seem to enjoy more certainty than foreign applications during the pendency period in both countries. And China seems to give preferential treatment to domestic applications while this phenomenon cannot be found in the U.S. Similarly, Liegsalz and Wagner [2013] study the grant lags of 443,533 SIPO (Chinese State Intellectual Property Office) patent applications from 1990 to 2002 by multivariate duration analysis, and they find that, although Chinese patent examination procedure basically fulfills the international standard, Chinese applicants achieved faster patent grants than their non-Chinese counterparts after controlling for other determinants of grant lags. de Rassenfosse and Raiteri [2016] also find similar results. Based on a study of half a million patent applications filed in SIPO, they find there is no or very weak evidence supporting the existence of domestic applicants' advantage. However, Chinese applications are four to seven percentage points more likely to be approved in the industries that are strategically important to China. For other patent offices, Webster et al. [2014] find EPO and JPO (Japanese Patent Office) are both more likely to favor domestic applications in areas which have high technological specialization in the domestic economy.

Table 7.1: The status summary of applications

	Granted	Failed	Expired	On process	Passing rate	In total
Domestic applications	232	255	18	17	47.6%	522
International applications	192	265	8	13	42%	478

7.3 Empirical Model and Data

7.3.1 Data

To study the possible reason(s) of domestic inventors advantage, I create a hand-coded dataset of patent applications. The dataset is collected from three different sources:

(i) USPTO Patent Application Database;

(ii) USPTO PAIR Bulk Database;

(iii) The Lens database.

I get the list of patent applications (1000) from the first database. To ensure that all the applications have already got their final examining results now, all the applications which I am using in this chapter were all published in 2010. This is because it usually takes at least 3 years for a patent application to be granted or get the final decision in the US. Even so, there are still 30 patent applications on the process of prosecution. To ensure the randomization of patent samples, I pick all the 796 applications published in January 2010, and another 204 applications from February based on the patent publication number order. In order to control the industry diversity, all the applications are from pharmaceutical industry (US Patent Classification No.: 424). This database also tells us whether the application is a domestic application or not. If at least one inventor is a domestic inventor, it is a domestic application, otherwise it is an international application.

The second database provides the examining results of all these applications. The status of all these patent applications include: Granted; Abandoned (failed); Expired due to nonpayment of maintenance fees[3]; On the process[4]. In all these 1000 applications,

[3]The patentee needs to pay a Patent Maintenance Fees of US Dollar 1,600, 3,600, 7,400, when her patent is due at 3.5 years, 7.5 years and 11.5 years respectively

[4]This includes still on appeal, or non final action mailed, or docketed new cases

522 are domestic applications and 478 are international applications. For the status, 424 were granted as patents and 510 failed, the remaining 66 are either expired or still on the process. For domestic applications, 232 of them were granted while 255 failed, and for international applications, 192 were granted while 265 failed. The passing rate is 47.6% for domestic applications and it is a bit higher than the passing rate of international applications (42%). The status summary is displayed in Table 7.1.

The third dataset is The Lens[5] which provides the full-text documents of patent applications. As the first project, I only keep the specifications and claims and remove general information and drawings, and I follow the parsing procedure of Loughran and McDonald [2011] to remove all tables as well. I also get the basic information about applications from Lens, including the number of forward citations and patent family size.

As with the previous two projects, I run the computational linguistic program to the whole document of all these 1000 patent applications to examine the readability level.

7.3.2 Empirical Model

We have already seen that domestic applications have a higher granted rate than international applications (47.6% vs 42%) in my dataset. But this simple comparison cannot ascertain the existence of domestic inventor advantage, it just gives us a good starting point to study the reasons behind this different granted rate. In order to have a detailed picture of my dataset before I go to the empirical model, I test whether there are any other differences on other factors like technical value and readability which may potentially affect the granted rate of patent applications as well.

I first do a simple comparison to check whether there is a significant difference between domestic and international applications on readability and technical value. Furthermore, I make this comparison just on granted patents and see whether the results change or not. (See the full results in Table 7.2)

For the technical value, I use patent family size as the value index as the first project,

[5]Please see the link of Lens:https://www.lens.org/lens/

Table 7.2: The results of comparison of technical value and readability

	Technical value (Family size)		Readability (Fog Index)	
	All applications	Granted applications	All applications	Granted applications
International applications	12.7	15.9	20.1	20.0
	(12.4)	(12.3)	(2.4)	(2.3)
Domestic applications	13.2	17.6	19.4	19.2
	(15.1)	(17.3)	(2.1)	(2.1)
P_value	0.025	0.47	0.00	0.00

and a simple ttest[6] is used to compare whether the patent value is same between domestic and international applications. The mean of patent family size is 13.2 for domestic applications and it is 4% higher than international applications (12.7). The difference is significant on 95% level of confidence. For the granted applications, although domestic applications still have larger average family size than international applications (17.6 vs 15.9), the different is not significant (p_value <0.47).

For the readability, I choose Fog as the index. The reasons why I use Fog Index as the readability index here are 1) as I discussed in the first project, Fog Index has been widely used in financial research; and 2) compared to the 106 features I use in the first and second project, I need a simple but effective index as a measure to control for the overall level of readability of each patent document. The mean of Fog Index is 20.1 for international applications and it is 3.6% higher than domestic applications (19.4). Because the higher of Fog index is, the more difficult of the text is, international applications are more difficult than domestic applications, and the difference is significant on 99% level of confidence. For the granted applications, the mean of Fog Index is 20.0 for international applications and it is 4.2% higher than domestic applications (19.2), and the difference is significant on 99% level of confidence as well. Here I also use the methodology in project one and project two (put all features in turns) to double check whether domestic applications are more readable than international applications. There are 78 features which is significantly different between domestic and international applications, and 51 of them (65.4%) show that domestic applications are more readable than international applications. [7].

[6]Although the patent family size is not normal distributed, we can still use ttest here, since we have relatively large sample size: 478 international patent applications and 522 domestic patent applications.

[7]I report the detailed results in Appendix E

From the results above, we can find that domestic applications are better than international applications in readability, and this result still holds in granted applications (significant on 99% level of confidence). For technical value, domestic applications are more valuable than international applications in general (significant at 95% level of confidence), but this significant difference no longer exists in granted applications.

Based on these findings, the natural questions that can be asked are: whether the different granted rate is related to technical value and readability, and whether domestic inventor advantage can still be found when controlling the readability and technical value of applications. Therefore, I will test the following questions in empirical models:

(i) Domestic applications are more readable than international applications;

(ii) Domestic applications are more valuable than international applications in terms of technical value;

(iii) Opportunistic enforcement of national treatment.

I start with a simple a regression analysis (probit) of the probability to test whether readability, technical value and domestic inventor alone can lead to higher granted rate (model 1, 2, 3).

$$y_{ij} = \alpha_{ij} + \beta_{1ij} * domestic + e_{ij}(1)$$

$$y_{ij} = \alpha_{ij} + \beta_{1ij} * familysize + e_{ij}(2)$$

$$y_{ij} = \alpha_{ij} + \beta_{1ij} * readability + e_{ij}(3)$$

$$y_{ij} = \alpha_{ij} + \beta_{1ij} * domestic + \beta_{2ij} * familysize + e_{ij}(4)$$

$$y_{ij} = \alpha_{ij} + \beta_{1ij} * domestic + \beta_{2ij} * readability + \beta_{3ij} * familysize + e_{ij} (5)$$

$$y_{ij} = \begin{cases} 1 \text{ if } y_{ij} > 0 \text{ (application is granted)} \\ 0 \text{ if } y_{ij} < 0 \text{ (application is refused)} \end{cases}$$

where readability is the value of Fog Index in our linguistic model; familysize is the patent family size which is the index of technical value of patent. "domestic" is a dummy variable, domestic = 1, when it is a domestic application and domestic = 0, when it is an international application.

I then add technical value index in Model 3 to control the technical value of applications and see whether domestic inventor advantage still exists or not (Model 4). I further add readability index to the Model 4, and study whether domestic inventor advantage can still be found in the new model(Model 5).

7.4 Results

In this section, I examine whether technical value and readability level of patent applications may affect the granted rate and the relationship between these two features and domestic inventor advantage.

In Table 7.3 and 7.4, I present the results of the empirical models. In model 1, domestic inventor alone significantly (at 90% level of confidence) increase the application's probability to be granted, since both the coefficient and marginal effect of domestic inventor(s) are positive and significant. The marginal effect of domestic inventor(s) indicates that having domestic inventor(s) would increase the granted rate by 5.6% (see Table 7.3).

The results from Model 2 and Model 3 tell the relationship between granted rate and technical value and readability(see Table 7.3). The positive coefficient and marginal effect of technical value means that applications with higher technical value are more likely to be

Table 7.3: The effect of technical value and readability

y = 1, when the application is granted, y = 0, when it is rejected

	Model 1		Model 2		Model 3	
	Coefficient	Marginal effect	Coefficient	Marginal effect	Coefficient	Marginal effect
Domestic	0.14*	0.056*				
	(0.08)	(0.03)				
Technical value			0.025***	0.01***		
			(0.003)	(0.001)		
Readability					-0.042**	-0.017**
					(0.019)	(0.07)

Standard error in brackets, [***] significant at 1 % level, [**] significant at 5 % level, [*] significant at 10 % level.

granted. The negative coefficient and marginal effect means that applications with higher readable language (the higher of the Fog Index is, the more difficult the application is.) are more likely to be granted. Given better technical value and readability level both can increase the granted rate, it is worth to test whether domestic inventor advantage can still be found or mitigated when controlling the technical value and readability of patent applications.

Table 7.4: The effect of domestic inventor(s)

y = 1, when the application is granted, y = 0, when it is rejected

	Model 1		Model 4		Model 5	
	Coefficient	Marginal effect	Coefficient	Marginal effect	Coefficient	Marginal effect
Domestic	0.14*	0.056*	0.15*	0.055*	0.11	0.043
	(0.08)	(0.03)	(0.08)	(0.03)	(0.08)	(0.03)
Technical value			0.026***	0.01***	0.026***	0.01***
			(0.003)	(0.001)	(0.003)	0.001
Readability					-0.045**	-0.017**
					(0.019)	(0.07)

Standard error in brackets, [***] significant at 1 % level, [**] significant at 5 % level, [*] significant at 10 % level.

In model 4, I add technical value into model 1 to test whether domestic inventor advantage still exists when controlling the technical value of patent applications. The coefficient and marginal effect of domestic inventor(s) almost don't change. Given the coefficient of marginal effect of technical value in model 4 are still significant, these

results can be interpreted to mean that higher technical value can lead to higher probability to be granted for patent applications in general, but controlling technical value cannot mitigate the domestic inventor advantage (i.e. when technical value is same, domestic applications still have 5.5% higher chance to be granted.). However, in model 5, after I add readability into model 4, both the coefficient and marginal effect of Domestic inventor(s) reduce (average marginal effect reduces from 5.6% to 4.3%) and both of them become insignificant, but the coefficients and marginal effects of technical value and readability are still significant. This indicates that domestic inventor advantage may attribute to better readability of domestic application, and this consists with the fact that domestic applications are more readable than international applications (in general and in granted applications).

7.5 Conclusion

This project examines whether readability level may affect the probability of patent applications to be granted. The empirical analysis is based on a dataset of 1000 patent applications which applied for in USPTO. Using these data, I compare the granted rate between domestic applications and international applications to check the existence of domestic inventor advantage and how readability level and technical value of patent applications may affect the granted rate as well. The main result is domestic applications do have a higher granted rate. However, technical value and readability level also affect the granted rate, and the domestic inventor advantage which is found in [Webster et al., 2014]'s study seems disappear after controlling the disclosure level.

One may argue that both domestic and international applications need to use local patent attorneys, so there should not be significant difference between domestic applications and international applications. Given that the patent attorney fee is really expensive (sometimes over US dollars 300,000), it makes no sense that local patent attorneys would offer such poor service. There is some sense in this argument, however, the translation problem still exists for a foreign application, and the communication between foreign inventors and local patent attorneys might exacerbate the problem.

Chapter 8

Conclusions

This thesis aims to explore and demonstrate the use of computational linguistic analysis to measure the readability of patent text. By using readability as a proxy for the extent of disclosure in patent documents, I introduce a new method to the quantitative economic analysis of patents, and generate preliminary research results which shed the light on the further studies.

In Chapter 5, I find that applicants with different incentives may choose different levels of disclosure for their patents. By comparing the readability of university and industry patents, I find that university patents have higher disclosure level, and this result is consistent with the difference of business model between universities (licensing) and corporations (in-house commercialization). The results from Chapter 5 support that the difference of readability of patent documents is not random, and this give us a safe starting point to do further study.

Chapter 6 studies whether disclosure quality of patent documents can be one of the considerations in patent acquisitions. My theoretic model suggests that comparing to PEs, NPEs are more likely to acquire less readable patents. However, I could not find that non-practising entities (patent trolls) choose to acquire less disclosing patents in the empirical result.

In Chapter 7, I further test the possible reason(s) of domestic inventor advantage. I measure the readability of domestic and international applications which applied to

USPTO in January and February of 2010 on pharmaceutical patents (US Patent Classification N0.: 424). I find that better readability can lead to higher granted rate, and the domestic inventor advantage which is found in Webster et al.'s study [2014] seems to disappear after controlling the disclosure level.

In this thesis, the focus on the disclosure aspect of patents is novel to the literature, and better understanding of the role of patent disclosure has the potential to provide new insights that can help to improve the effectiveness of the patent system not only to stimulate innovation but also to make sure that this knowledge generates the largest possible impact in terms of economic growth.

Of course, since this is the first time that computational linguistic analysis is being used for patent documents, together with the limited time and resource, there are still some limitations in this thesis. First, this thesis might suffer from the imperfect of linguistic analysis model. In the first project, I have already checked how linguistic model works in patent documents. In general, it works pretty good, but there are still some problems: 1) the problem of overlapping function. There are 106 features in the model and some of them, like Fog formula, Kincaid formula and Flesch formula all based on the combination of sentence length and word length[1]. This will weaken the effectiveness of comparison results, since the results of group comparison (universities vs practising firms, NPEs vs PEs and domestic applications vs international applications) are based on the number of features. As the next step of this research, this question needs to be addressed. 2) whether all these features shall have the same weight and whether we can get a single readability index based on this linguistic model (or which feature in this model can be used as the unique readability index which grasp the characteristics of patent documents). Apart from group comparison, a single readability index is needed when we try to study the effect of readability, just like how Fog index is used in finance and accounting. It is worth to further test which feature is best specified in patent documents.

Second, the result of this thesis might also suffer from the relative smaller sample size, especially the second project, it only has 200 samples in total (100 each for NPEs

[1]The correlation between Fog index and Kincaid index is 0.97.

and PEs). Although a lot of effort has been put on data randomization to prevent the selection bias, a larger size sample (i.e. whole industry data within certain period) will be more persuasive, and cross-industries research may also be needed to consolidate the results.

Third, more control variables are needed to empirically explore the determinants that might affect the disclosure level of patent documents. For example, for all the practising firms, the firm size might be correlated with disclosure levels firms will choose in their patent documents. In their study, Veer and Jell points out that small firms may also focus on licensing their patents, and the motive of licensing loses importance as the company size increases. And for forward citations, as discussed in the first project, since a large part of forward citations are added by patent examiners, the results may suffer from the bias brought from using pooled citations. All these variables will be added as the next step.

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